

IESTI01 – TinyML

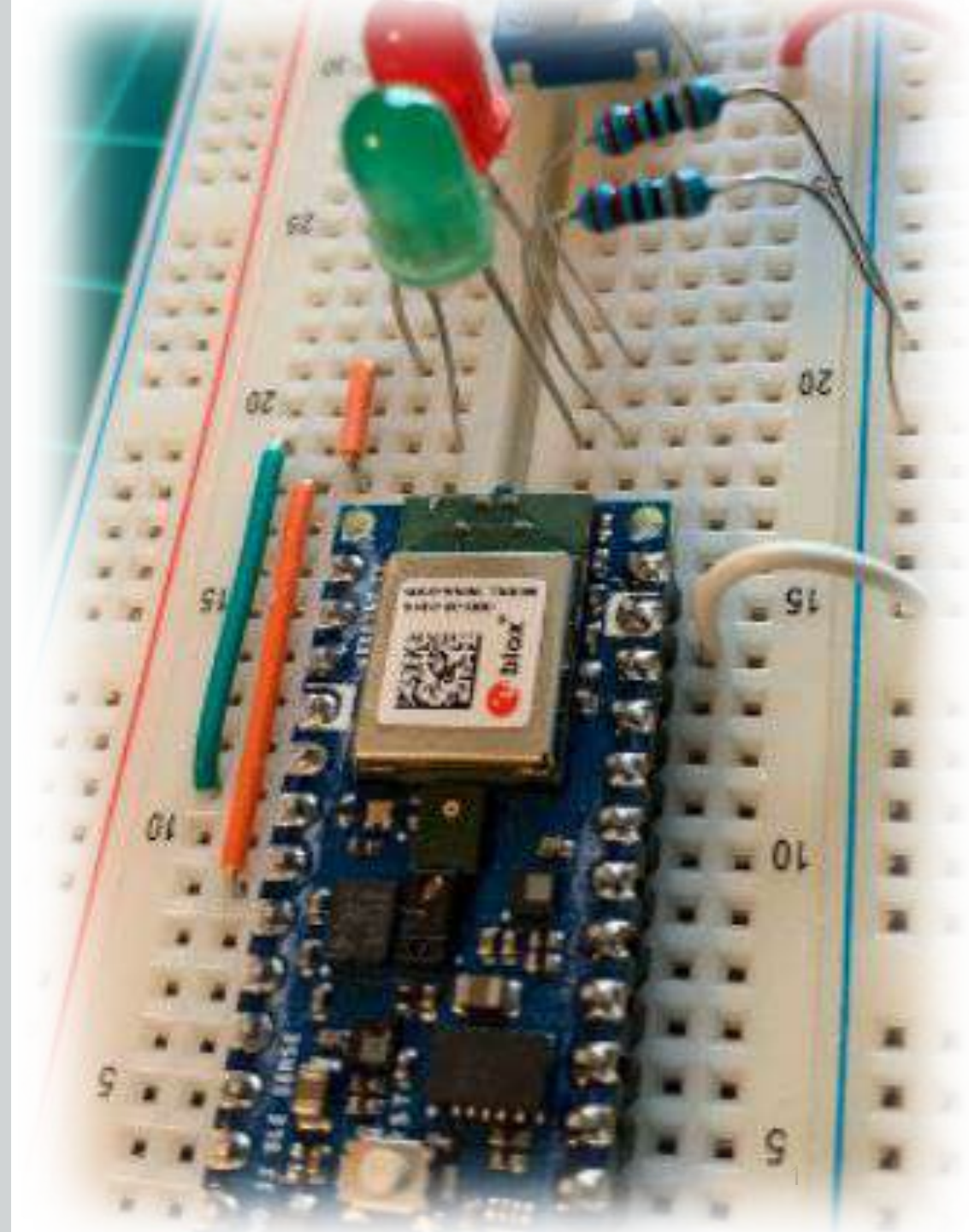
Embedded Machine Learning

1. About the Course & Syllabus



Prof. Marcelo Rovai

UNIFEI



Dear students,

Welcome to the first class of the IESTI01 (TinyML) EAD course. I am Professor Marcelo Rovai, a former student here at UNIFEI, and I am pleased to be with you this semester.

IESTI01 is a discipline that mixes Machine Learning (part of Artificial Intelligence) with small devices, such as microcontrollers and sensors, whose main characteristics are ultra-low power consumption, usually 32-bit CPUs, and a few kilobytes of memory.

We understand that the explosive growth of the Machine Learning field, the ease of use of software development platforms such as TensorFlow (TF), based on the Python language, and the current generation of powerful microcontrollers, make TinyML an indispensable topic of study for Engineering students in the areas of Electronics, as well as Computing and Control & Automation.

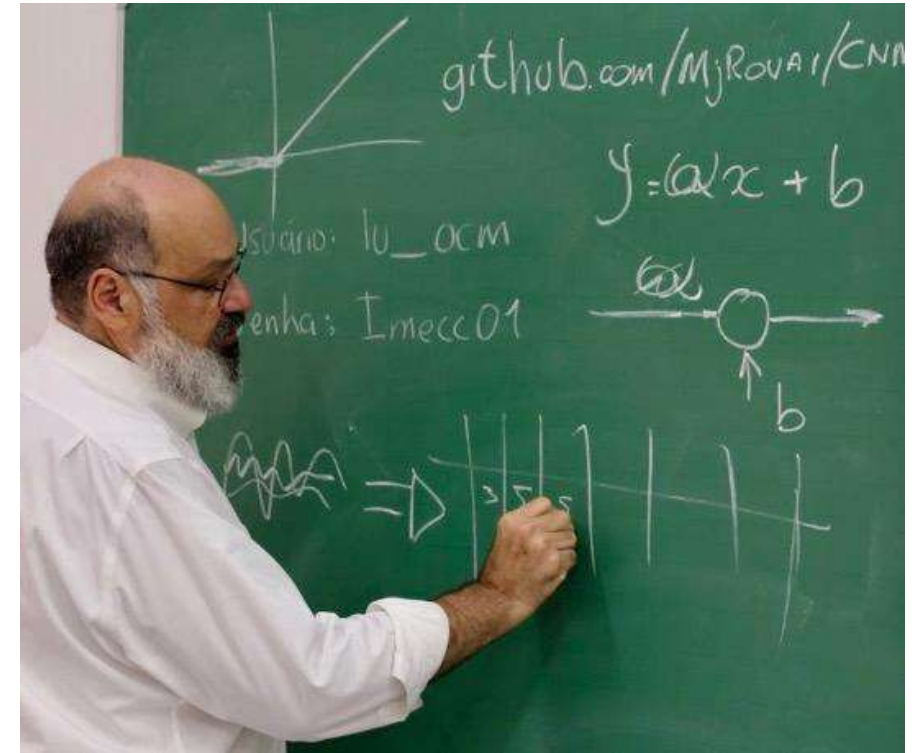
This mix of expertise and the pioneering nature of this discipline (We were the second university worldwide to have this type of course, Harvard School of Engineering being the first) leads us to significant challenges concerning the necessary basis for the minimum understanding of the matter. Thus, the time we have available for the course is short. So, We must commit ourselves to following the weekly classes and doing complementary activities such as readings, laboratories, and assignments. You can count on me to clarify doubts and to review the necessary concepts for a good understanding of the course (For that, send me an email or use the FORUM, available in each class).

Greetings from the south of the world!

Prof. Marcelo Rovai

Who I am

- Brazilian from São Paulo, **Data Science Master's degree by UDD**, Chile, and **MBA by IBMEC (INSPER)**, Brazil.
- Graduated in 1982 as an **Engineer from UNIFEI** with **Specialization from Poli/USP**, both in Brazil.
- Worked as a **teacher, engineer, and executive** in several technology companies such as CDT/ETEP, AVIBRAS Aeroespacial, SID Informática, ATT-GIS, NCR, DELL, COMPAQ (HP), and more recently at IGT as a Regional VP, and a Senior Advisor for Latin America.
- **Write about electronics**, publishing in sites as MJRoBot.org (Editor/Writer), Hackster.io (#1 Contributor), Instructables.com, and Medium.com (TDS – Towards Data Science).
- **Volunteer Professor** at UNIFEI (IESTI), teaching “Machine Learning applied to Embedded Devices” course (IESTI01) and EdgeAI Engineering (IESTI05).
- **TinyML4D group** and the **EDGE AIP** – the Academia-Industry Partnership of **EDGEAI Foundation** Co-Chair, promoting EdgeAI education in developing countries, underscoring a commitment to leveraging technology for societal advancement.



Marcelo Rovai

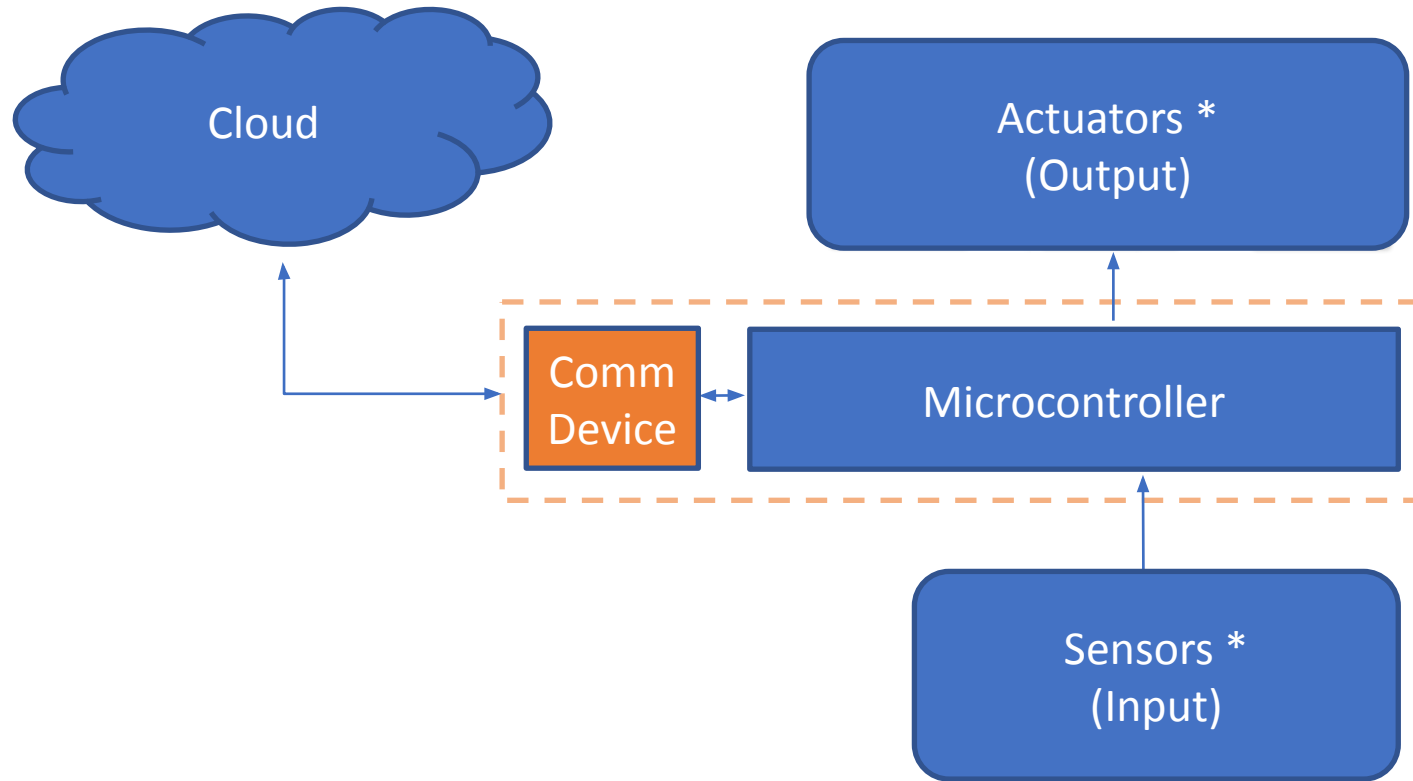
“**Edge AI** is a truly complete technology. As a topic, it makes use of knowledge from everything from the physical properties of semiconductor electronics all the way up to the engineering of high-level architectures that span devices and the cloud. It demands expertise in the most cutting-edge approaches to artificial intelligence and machine learning along with the most venerable skills of bare-metal embedded software engineering. It makes use of the entire history of computer science and electrical engineering, laid out end to end.”



Situnayake, Daniel; Plunkett, Jenny
AI at the Edge (pp. 215-216)
O'Reilly Media

Internet of Things (IoT)

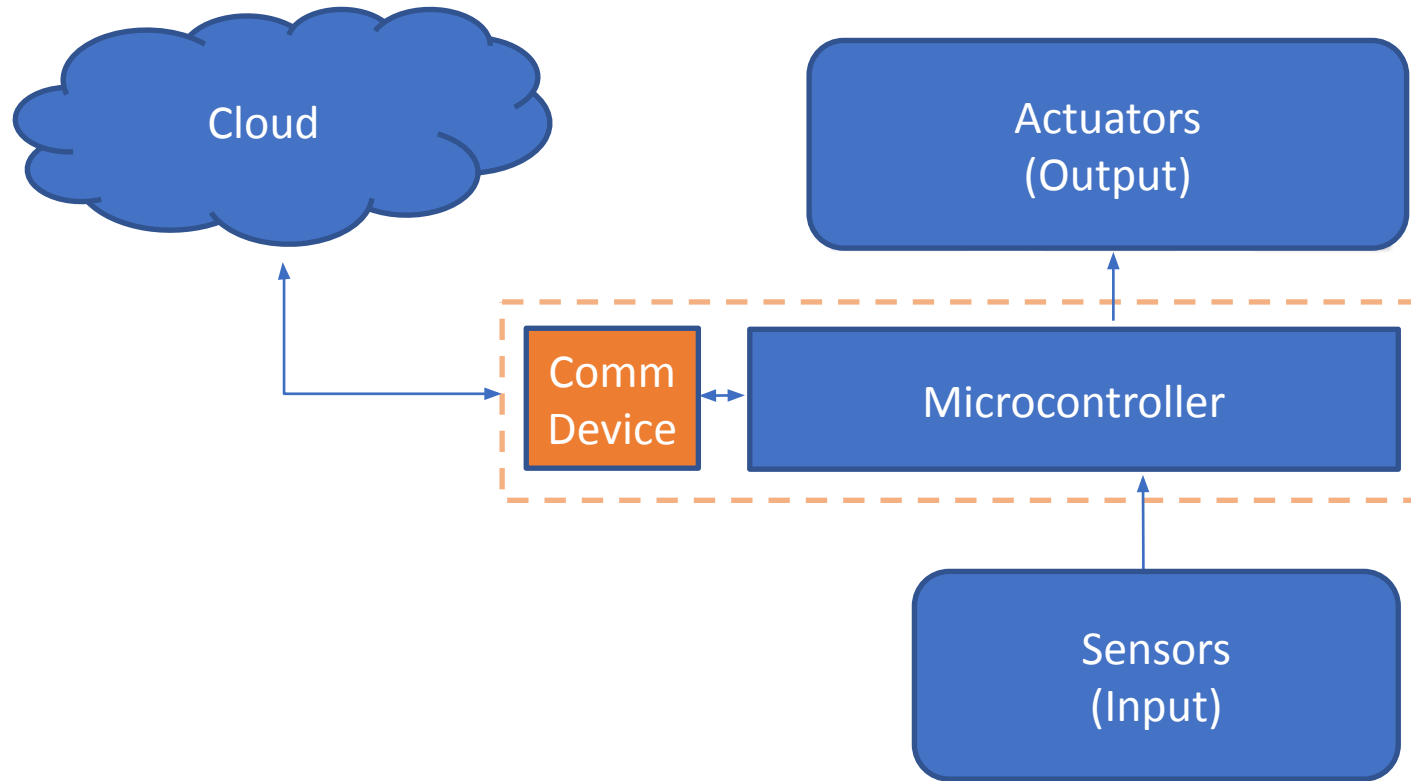
Typical IoT Project



* "Things"



Typical IoT Project



5 Quintillion

bytes of data produced
every day by IoT

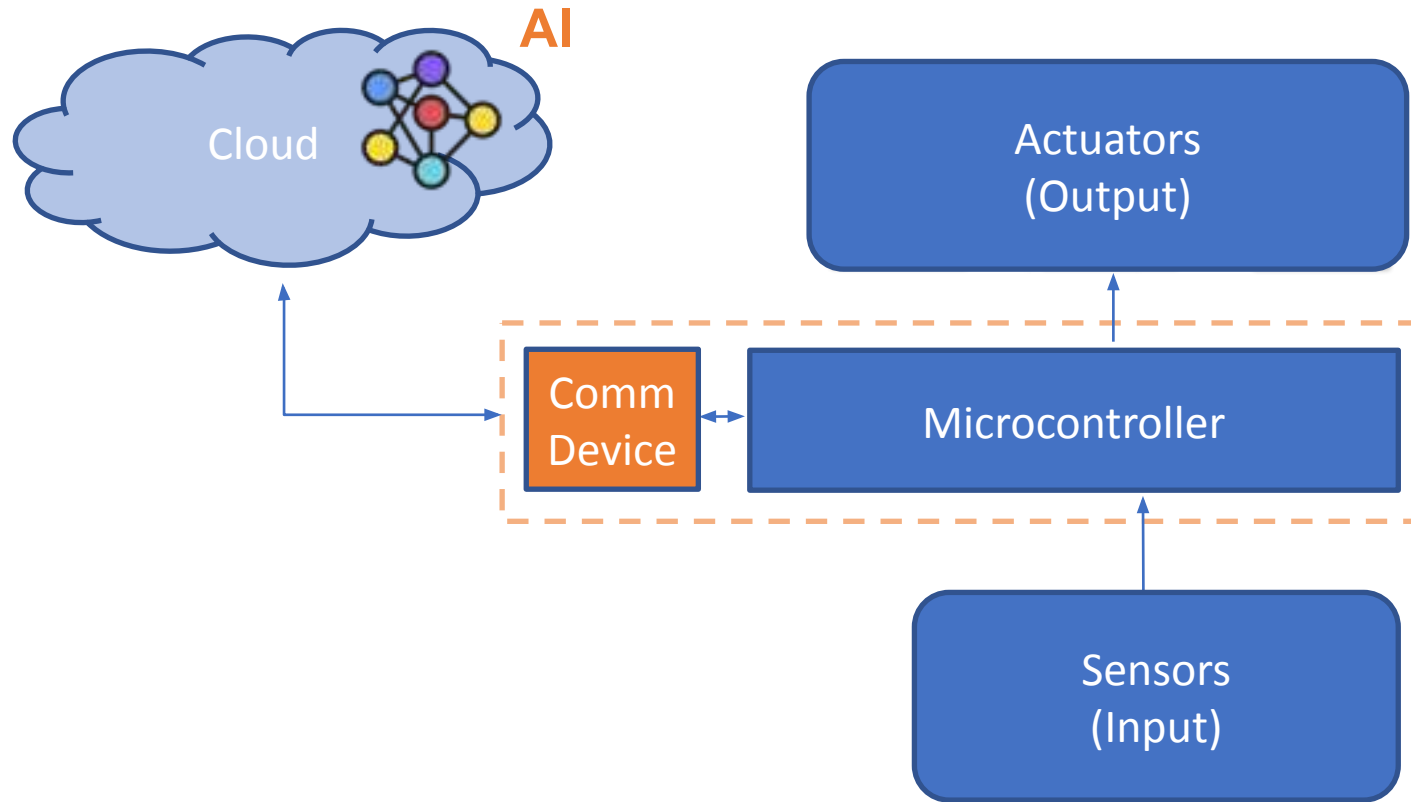
<1%

of unstructured data is
analyzed or used at all

Source: Harvard Business Review, [What's Your Data Strategy?](#), April 18, 2017

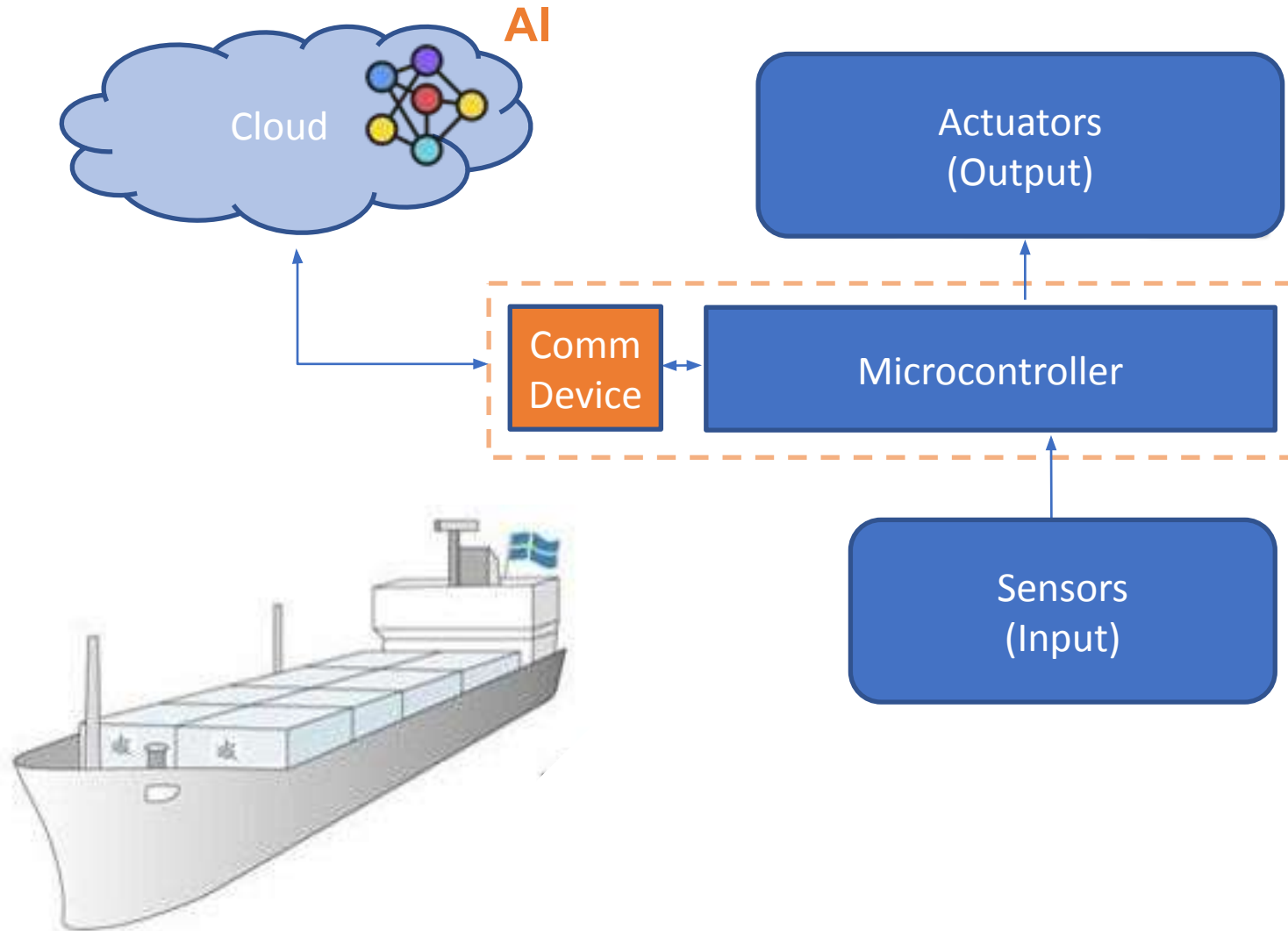
Cisco, [Internet of Things \(IoT\) Data Continues to Explode Exponentially. Who Is Using That Data and How?](#), Feb 5, 2018

Typical **A**IoT Project



Typical **A**IoT Project ...

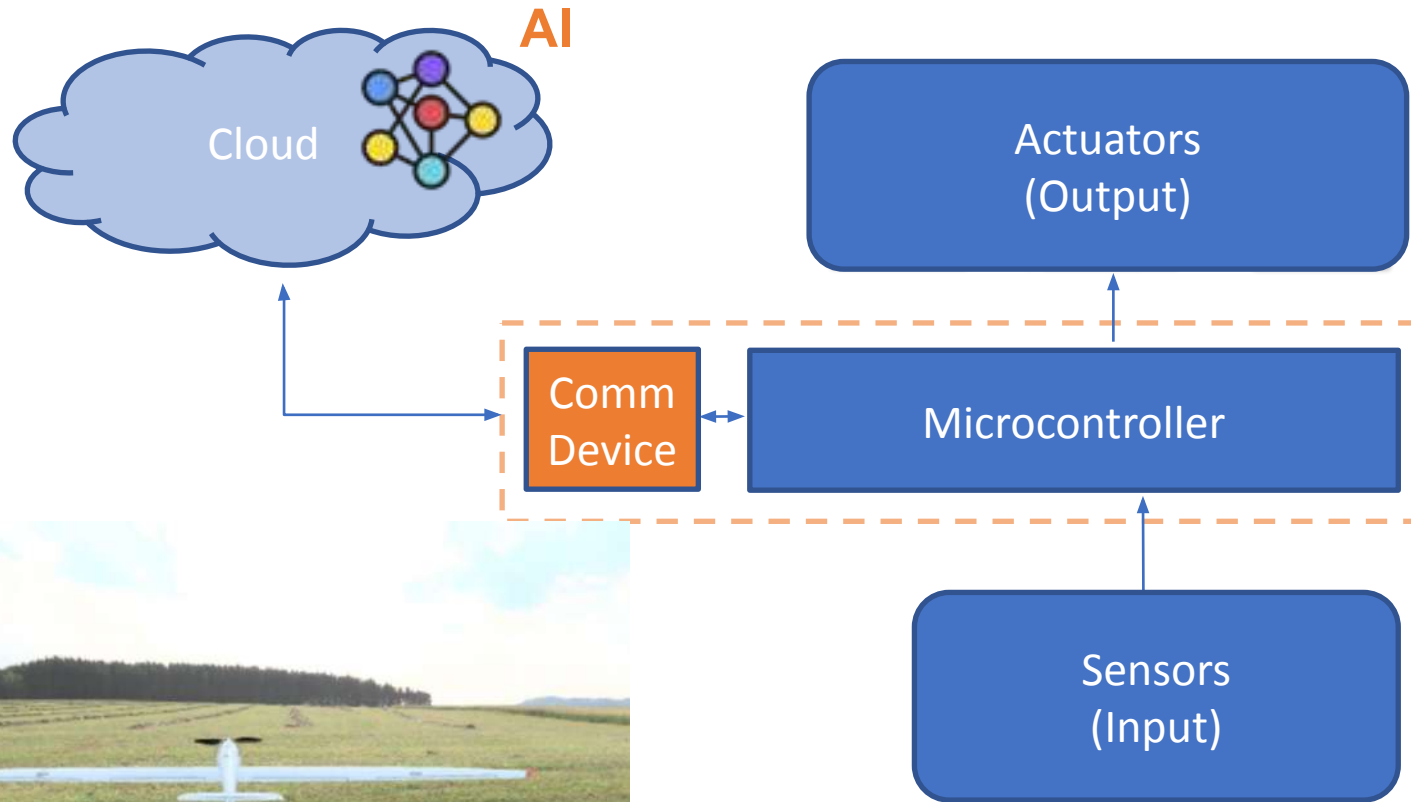
... **Issues**



Bandwidth

Typical **AIoT** Project ...

... **Issues**

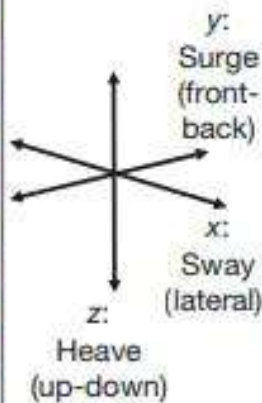
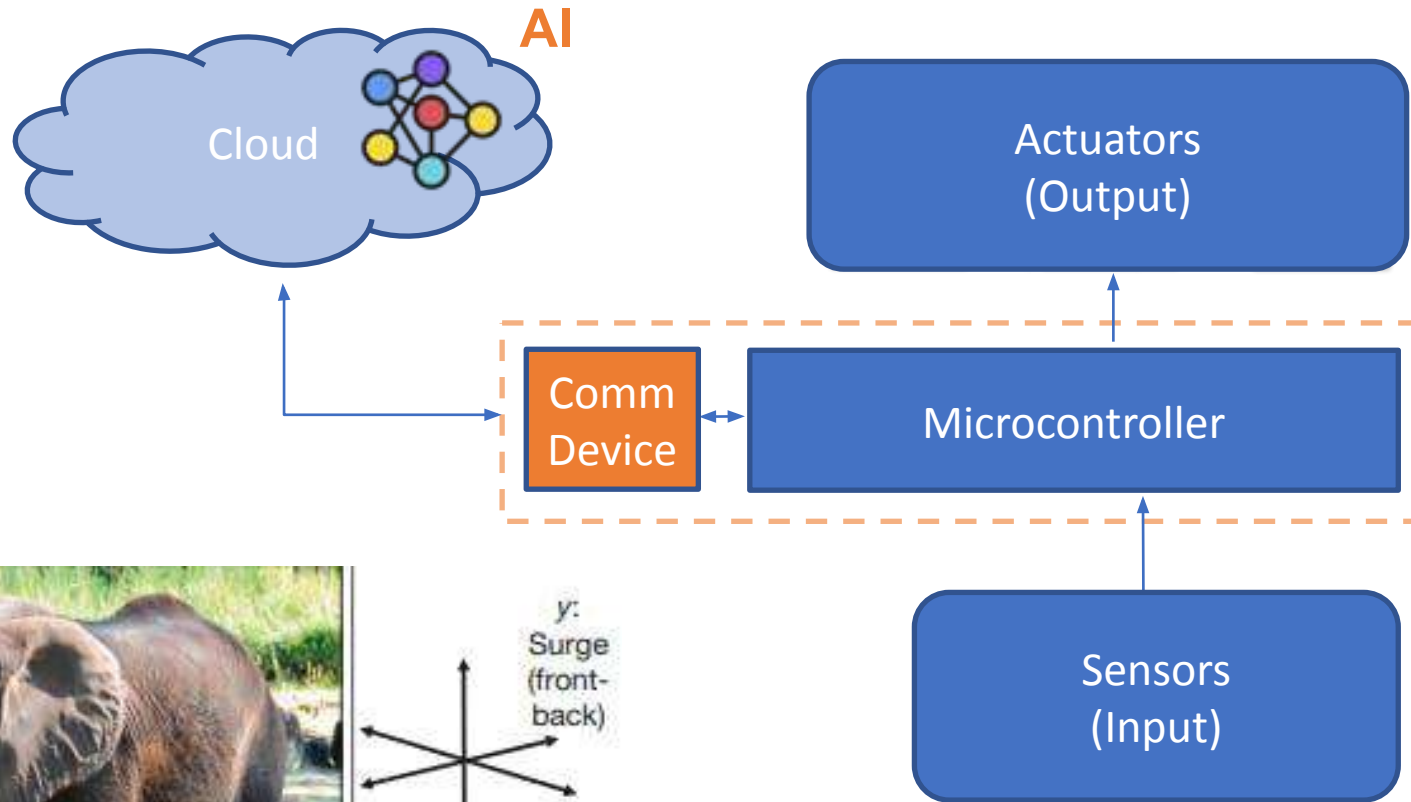


Bandwidth
Latency



Typical **A**IoT Project ...

... **Issues**

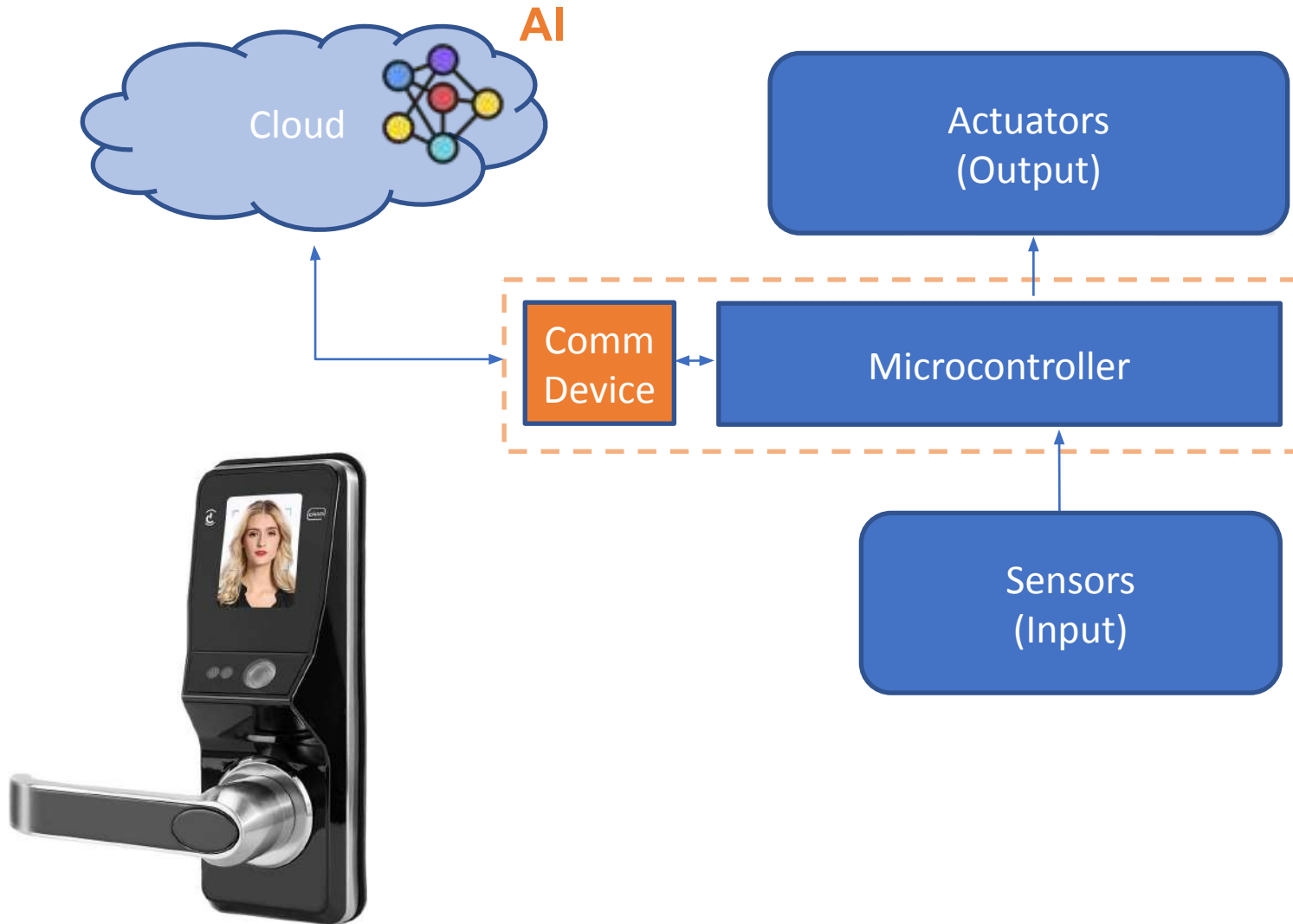


Bandwidth

Latency

Energy

Typical **A**IoT Project ...



... **I**ssues

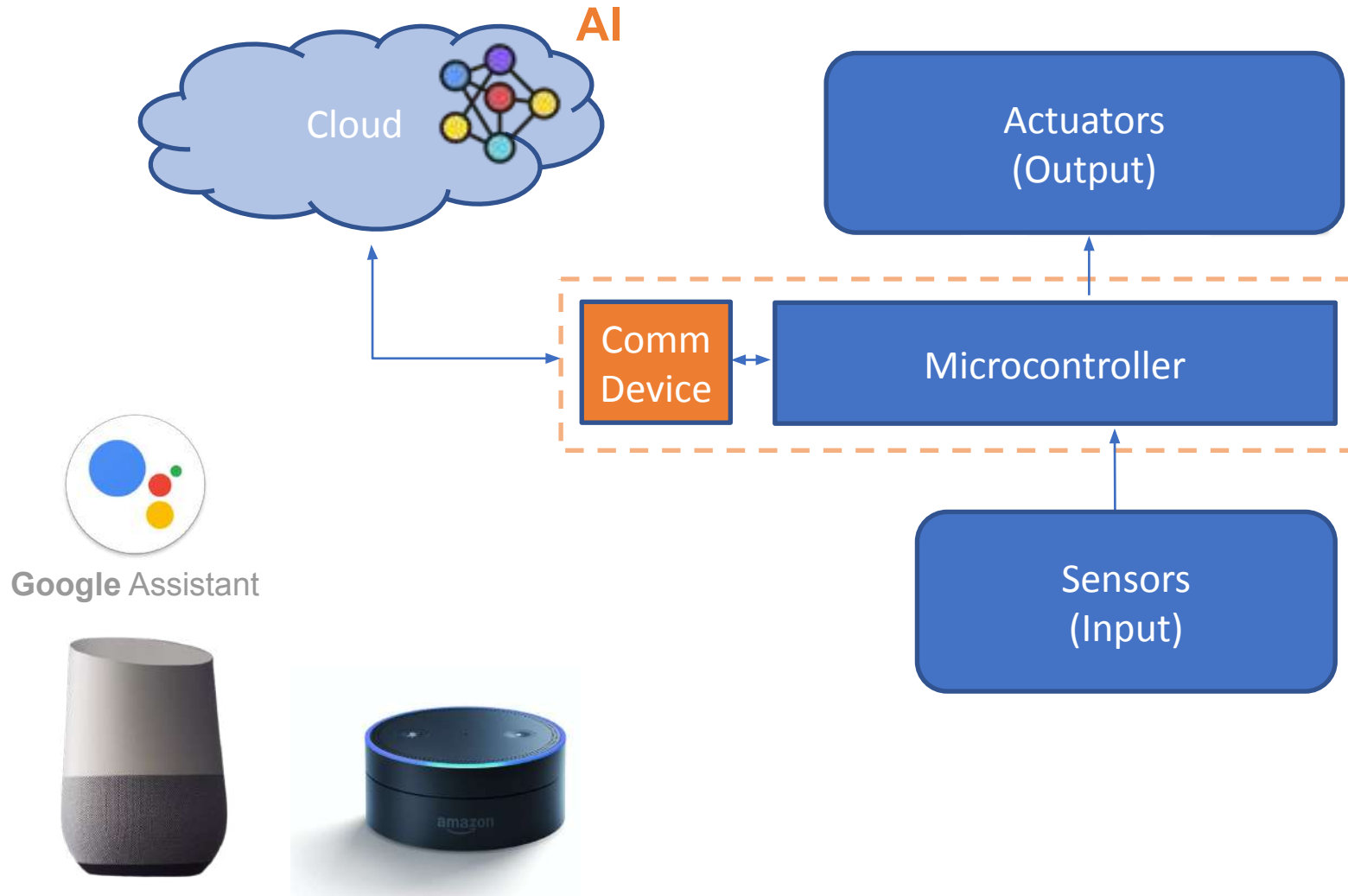
Bandwidth

Latency

Energy

Reliability

Typical **A**IoT Project ...



... **I**ssues

Bandwidth

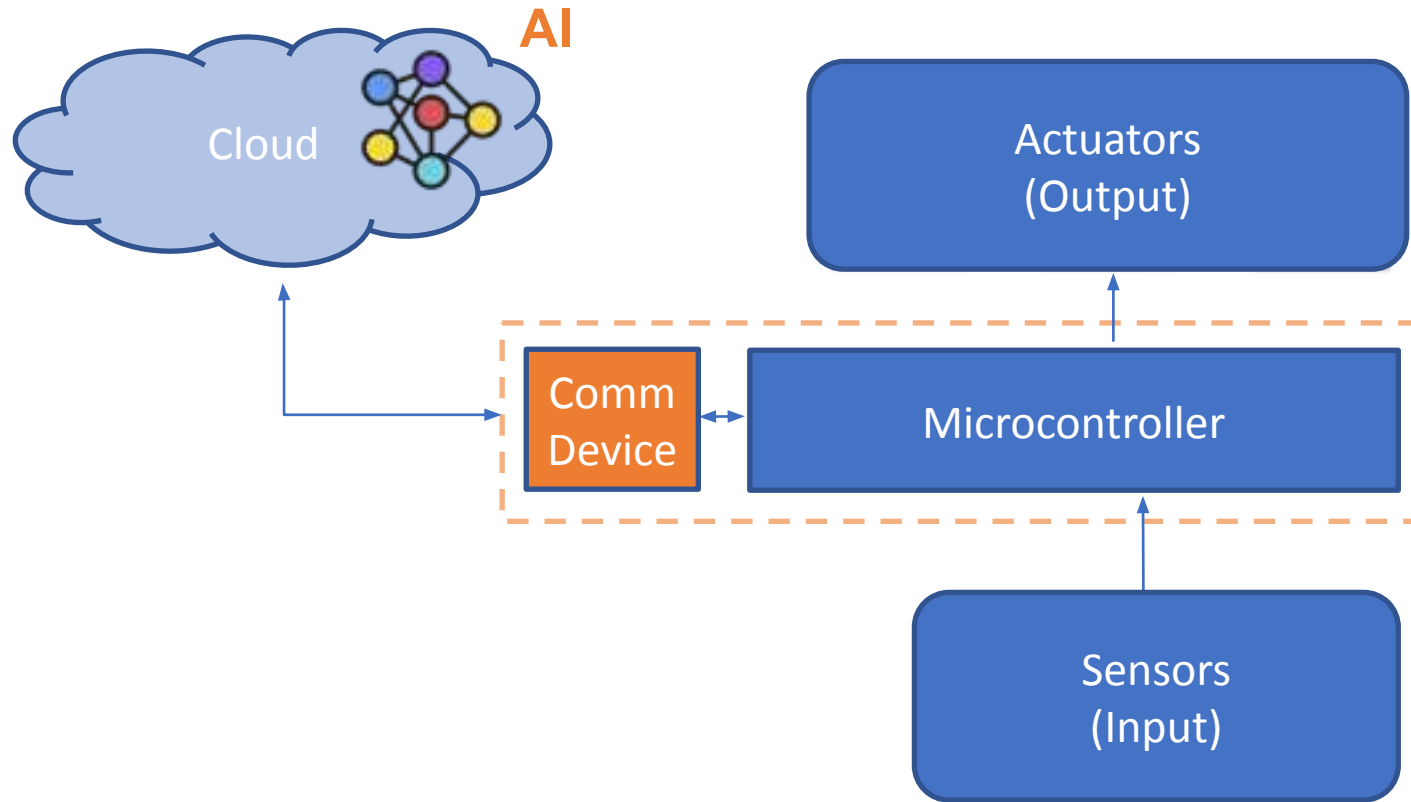
Latency

Energy

Reliability

Privacy

Typical **A**IoT Project ...



... **Issues**

Bandwidth

Latency

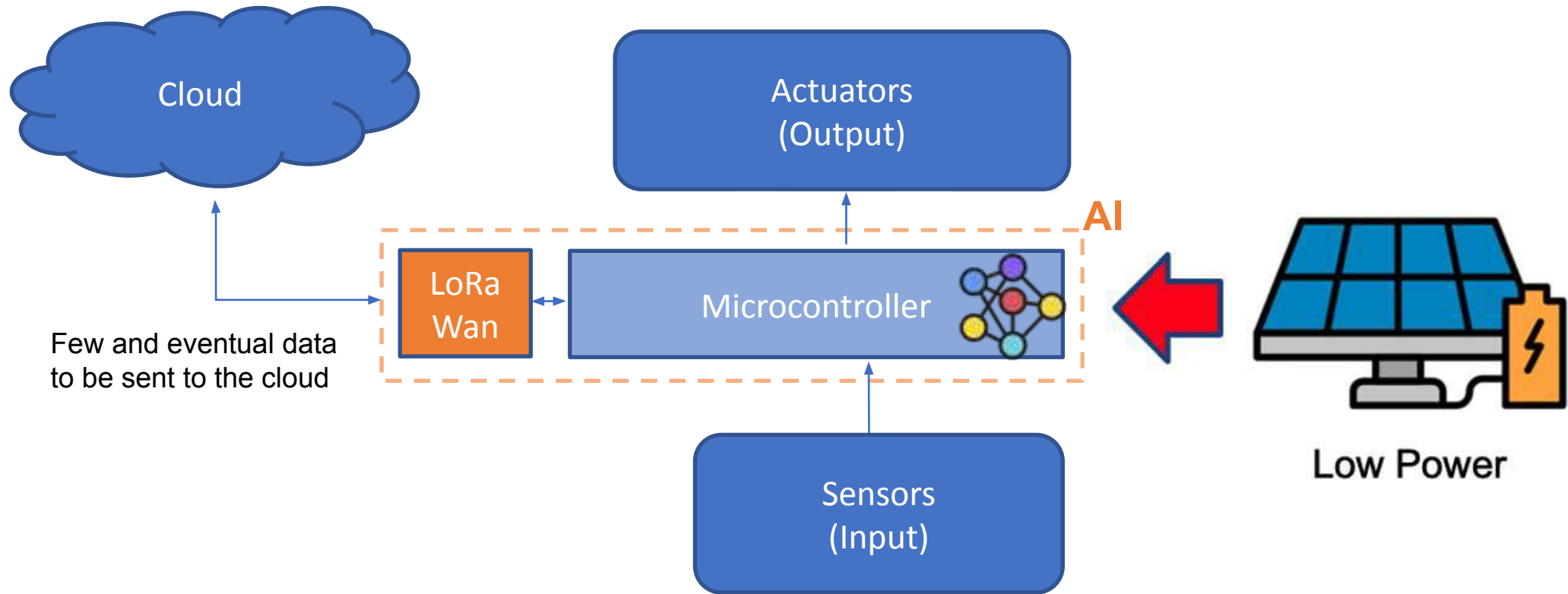
Energy

Reliability

Privacy

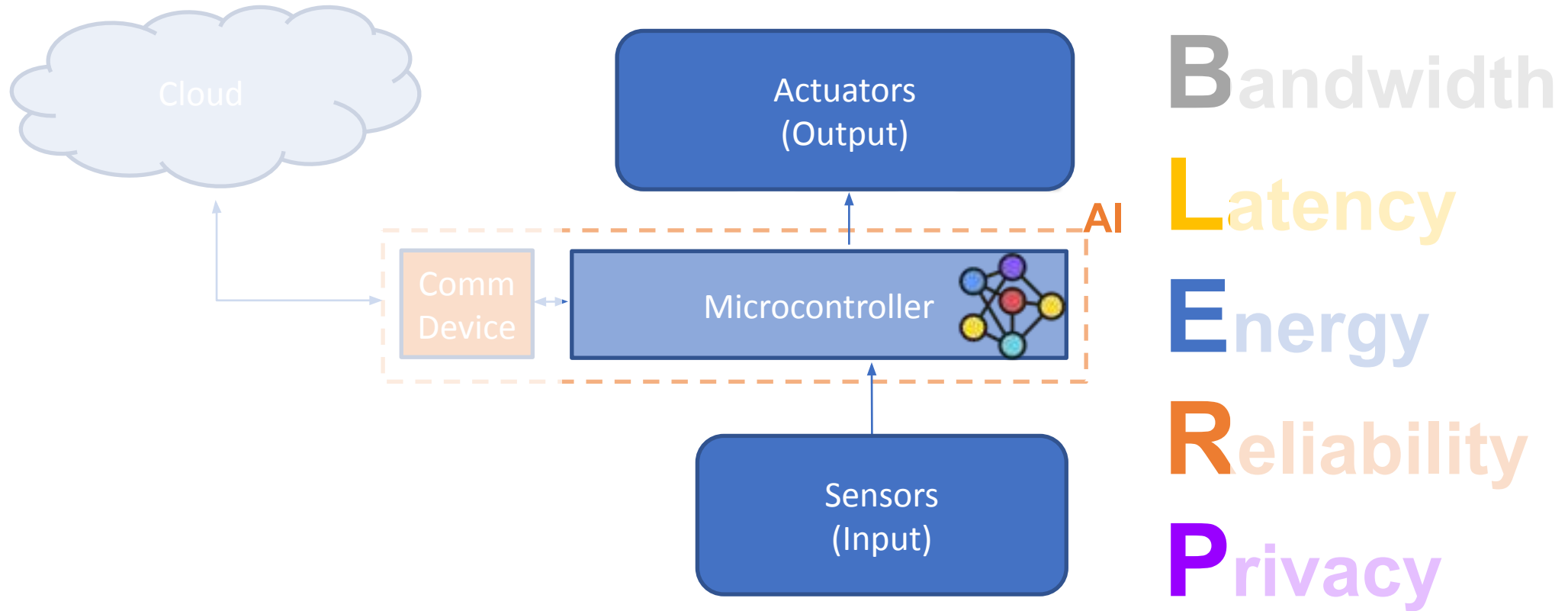
... **Solution ?**

IoT 2.0 * – Edge AI/ML * Intelligence of Things



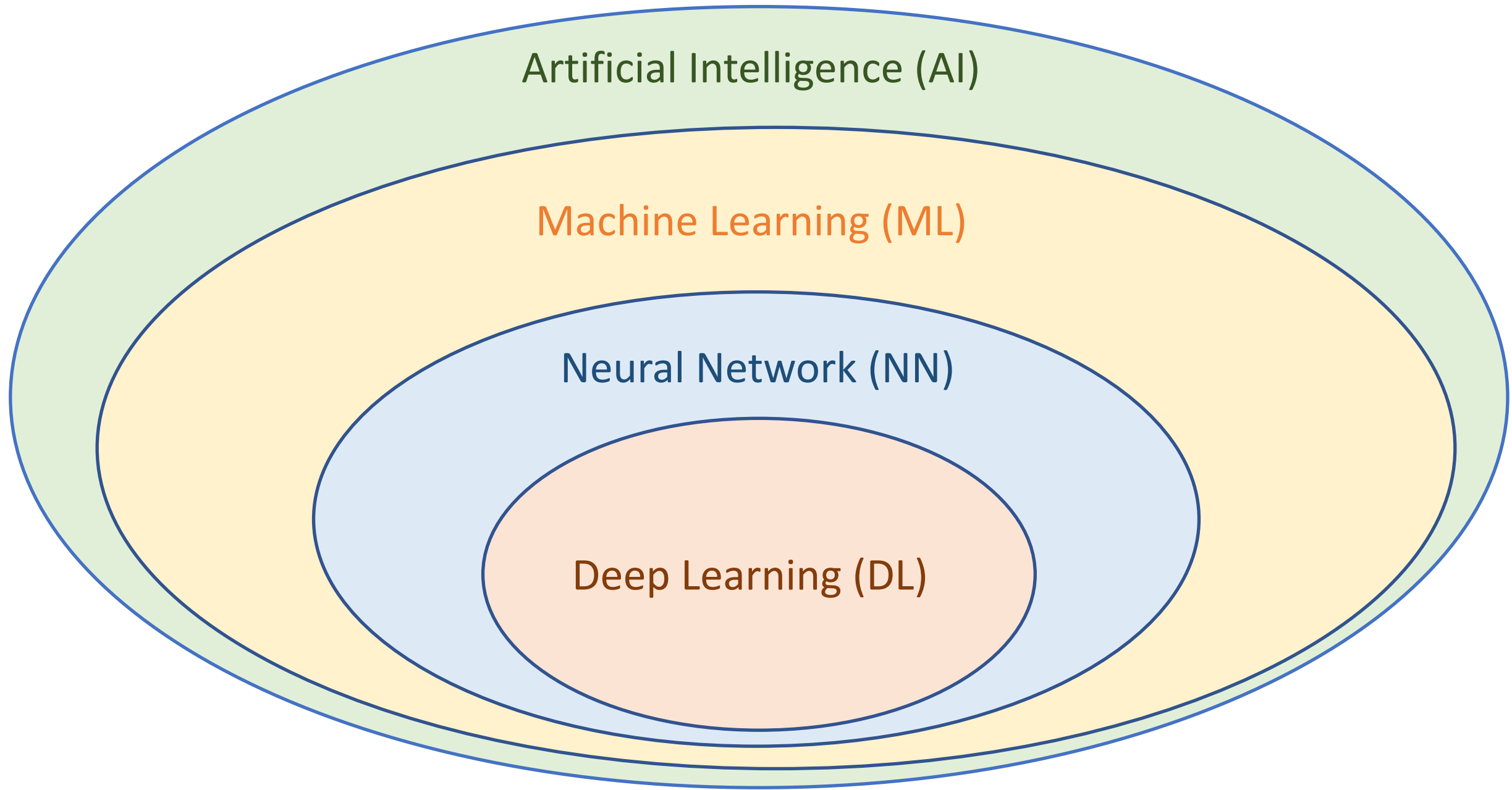
... **Solution** -> ML goes close to data

When to use an Edge AI/ML approach:

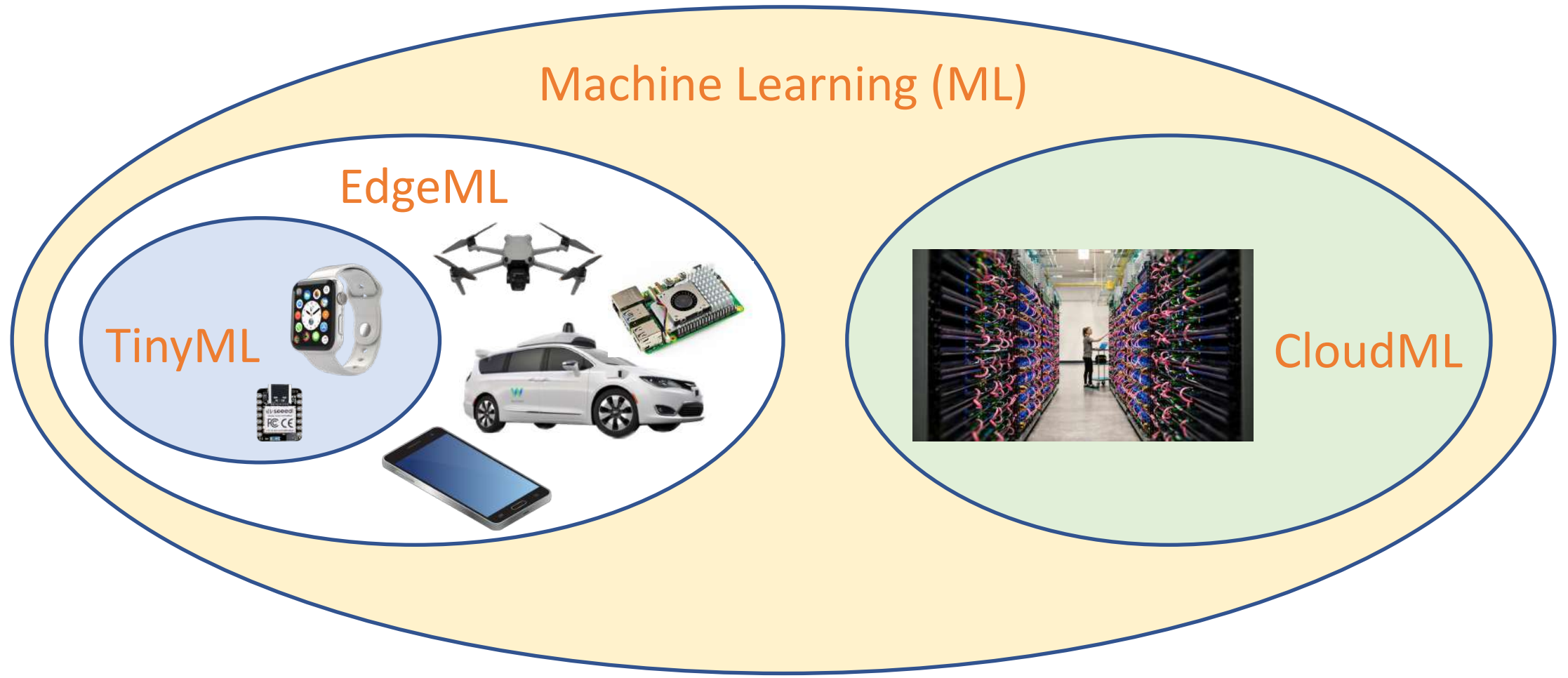


Embedded ML (TinyML)

Introduction



From Tiny to Cloud: One AI, Many Scales



From Tiny to Cloud: One AI, Many Scales



TinyML

20KB-1MB

Microcontroller

1W-

\$5-\$30



EdgeML

0.5GB-64GB

SBC/Phone

1-60W

\$30-\$900



CloudML

10GB-TBs

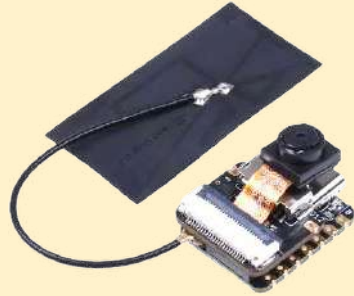
GPU/Server

500W+

\$10K-\$1M

Sustainability Matters

0.5W



0.4W



15W



20W+



100W+



100x more energy efficient than cloud AI

What is Tiny Machine Learning (**TinyML**)?

TinyML

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TinyML

Fastest-growing field of **ML**



What is Tiny Machine Learning (**TinyML**)?

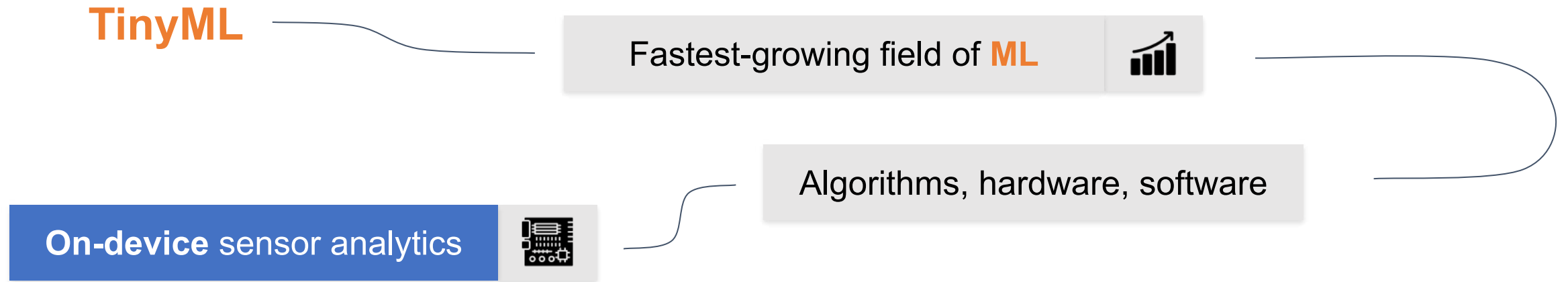
TinyML

Fastest-growing field of **ML**

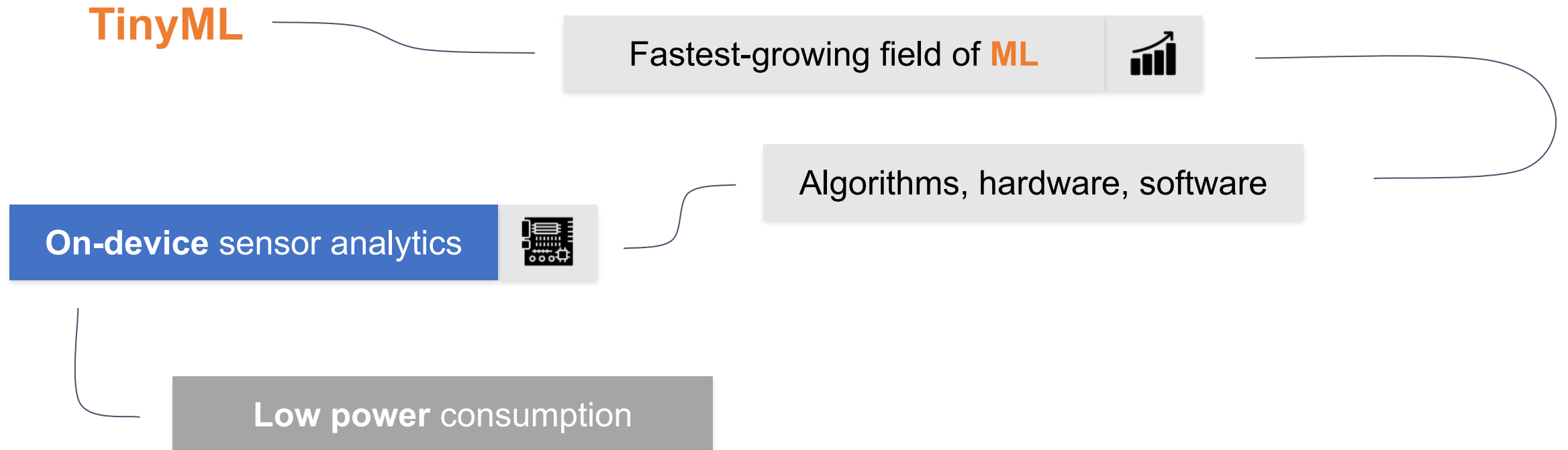


Algorithms, hardware, software

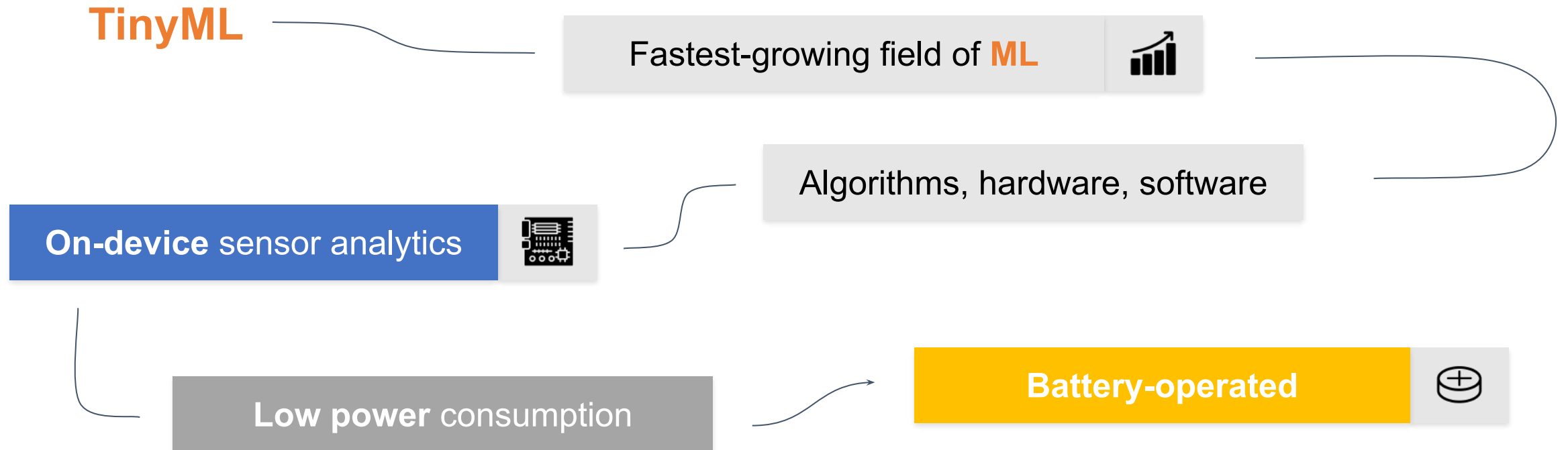
What is Tiny Machine Learning (**TinyML**)?



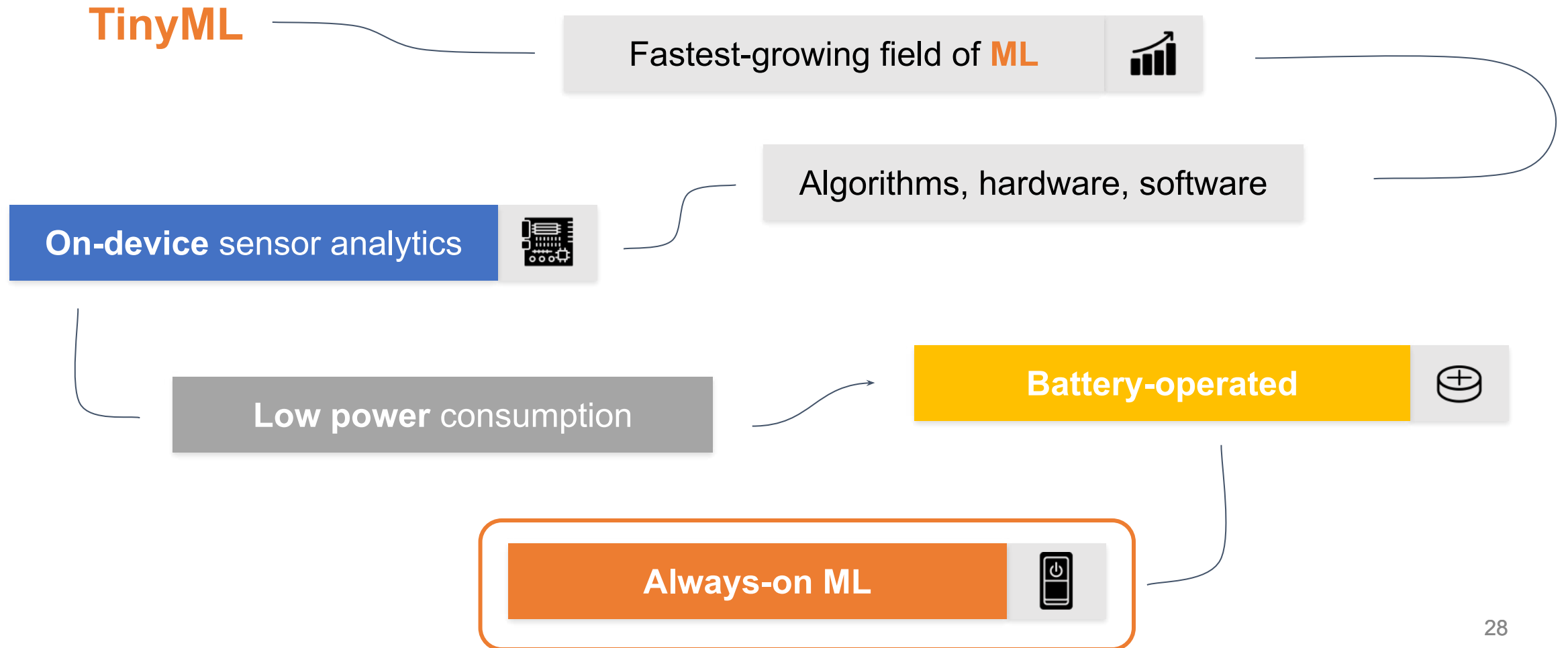
What is Tiny Machine Learning (**TinyML**)?

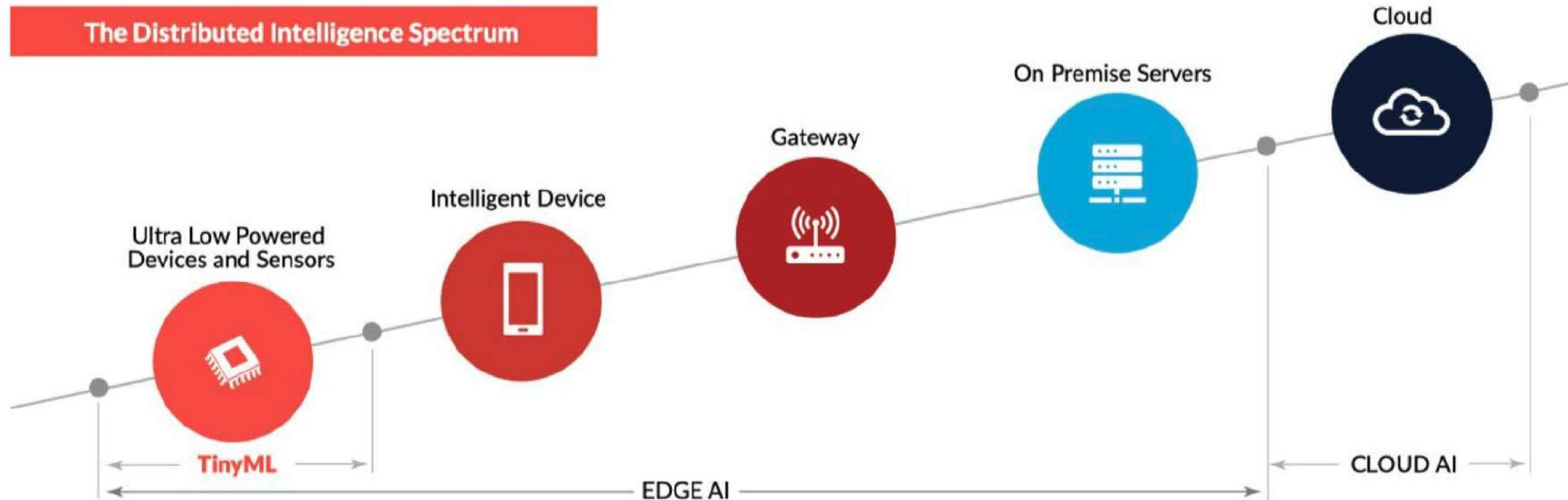


What is Tiny Machine Learning (**TinyML**)?

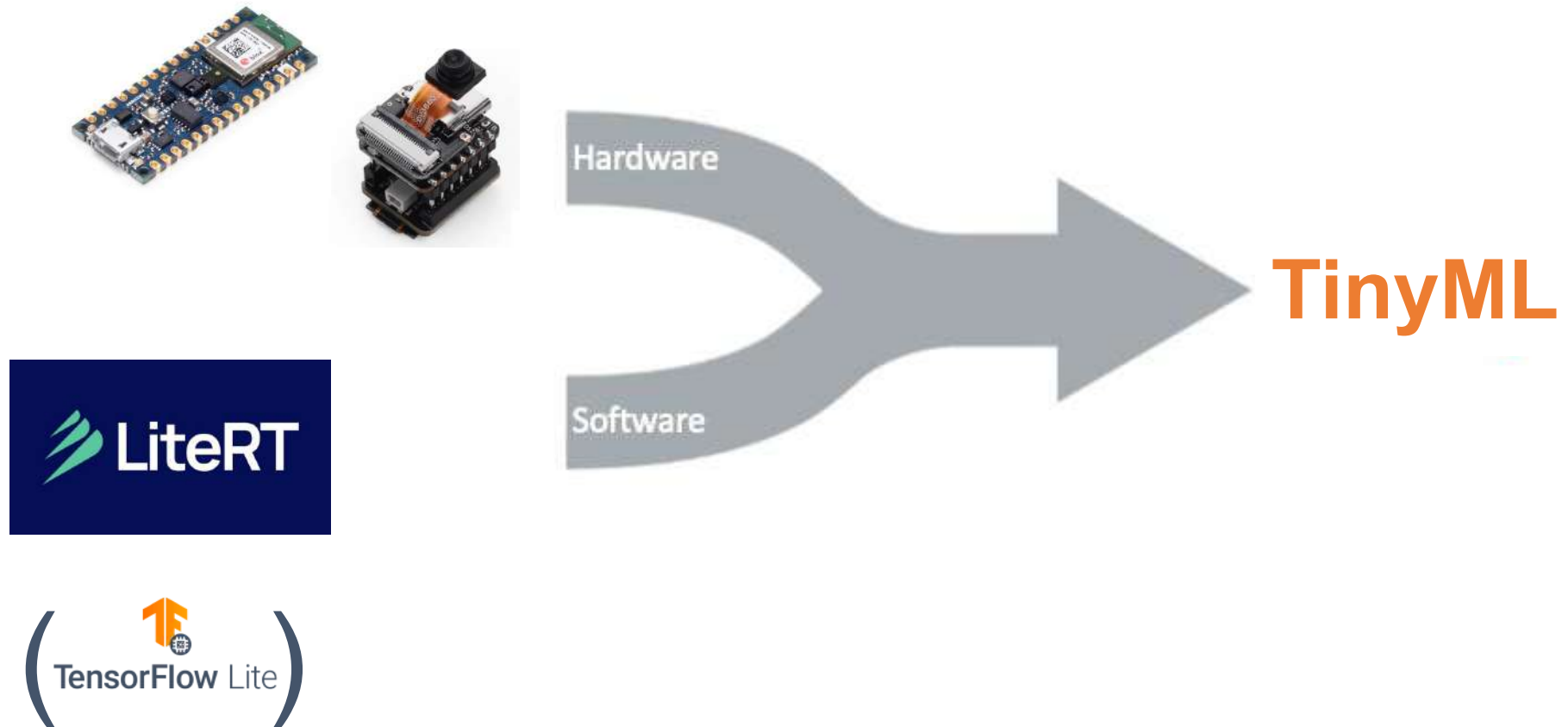


What is Tiny Machine Learning (**TinyML**)?



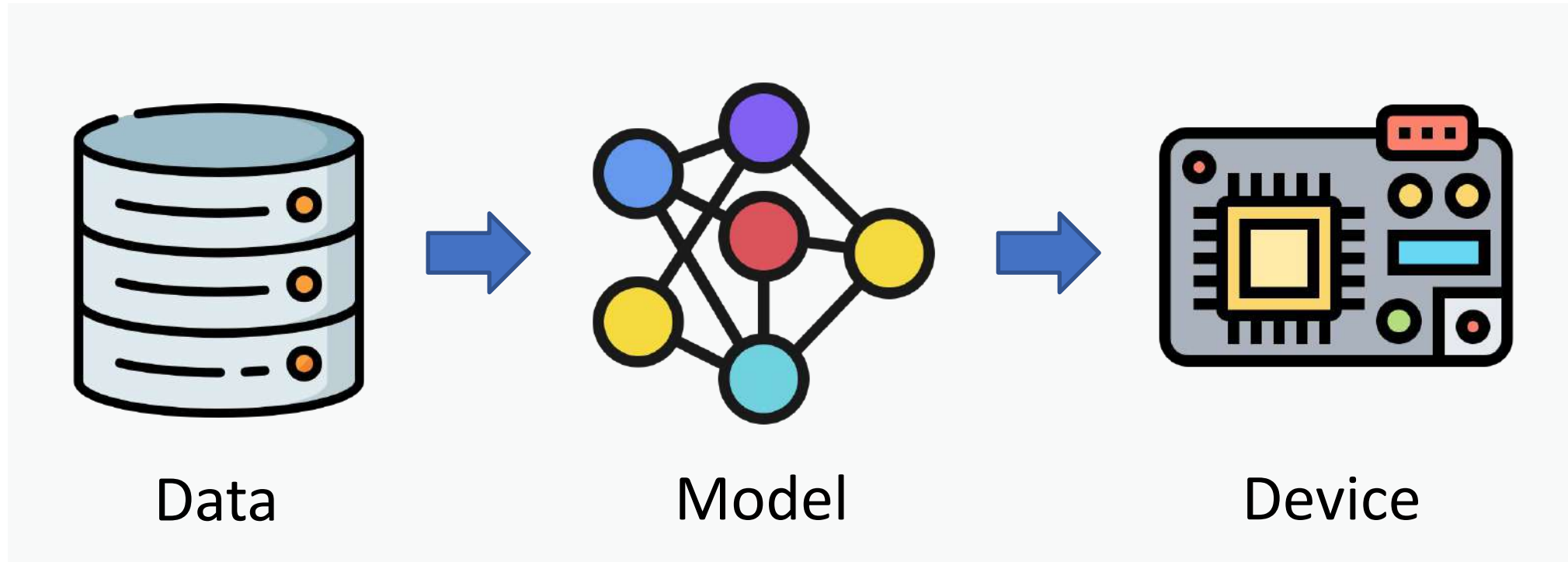


What Makes **TinyML** ?

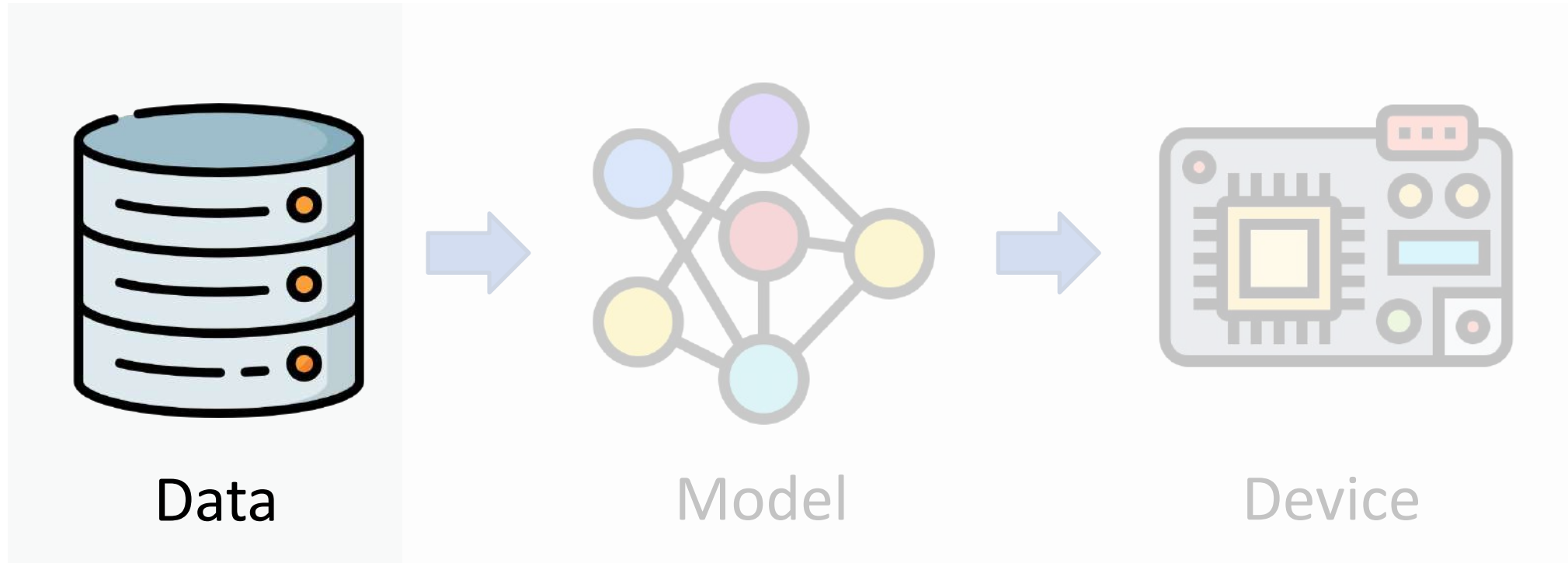


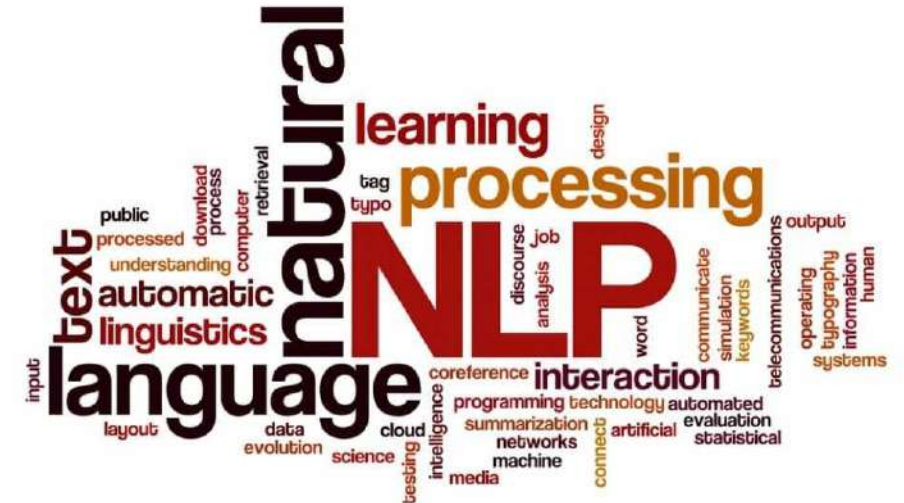
Challenges

ML Deployment Pipeline

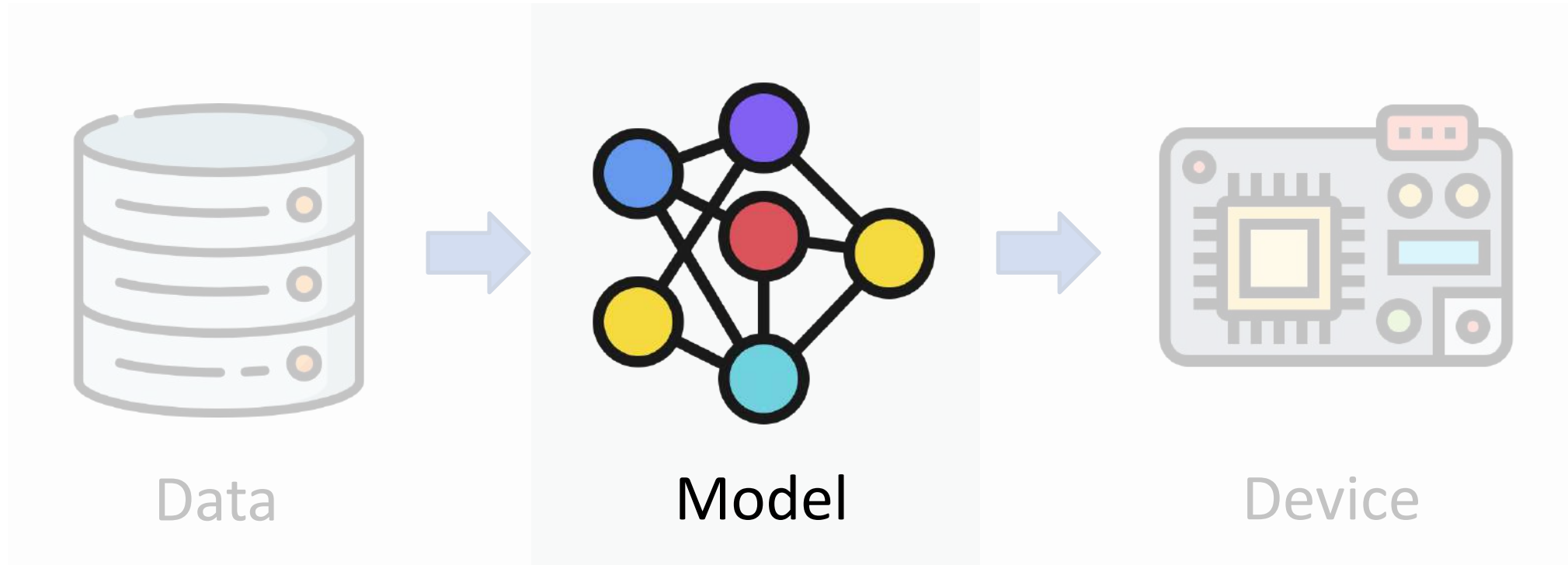


ML Deployment Pipeline



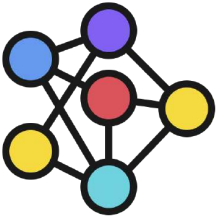


ML Deployment Pipeline





Neural Network Architectures



Vibration
Analysis

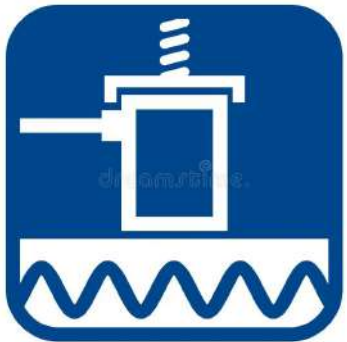


Image
Classification



Text
Generation



Image
Generation



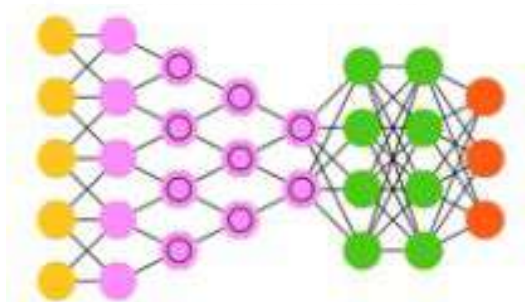
Large Language
Models- LLMs



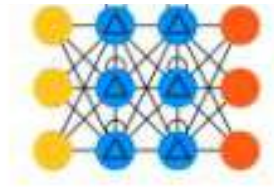
MLP - Deep Neural Network



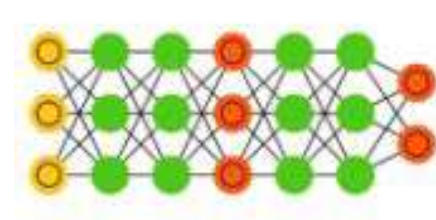
CNN - Convolutional NN



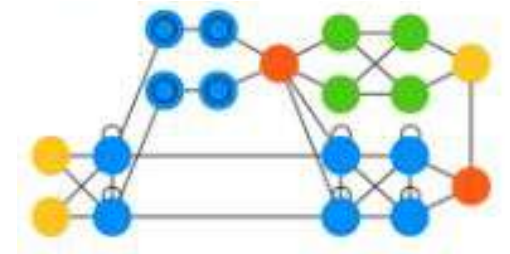
RNN - Recurrent NN (GRU/LSTM)



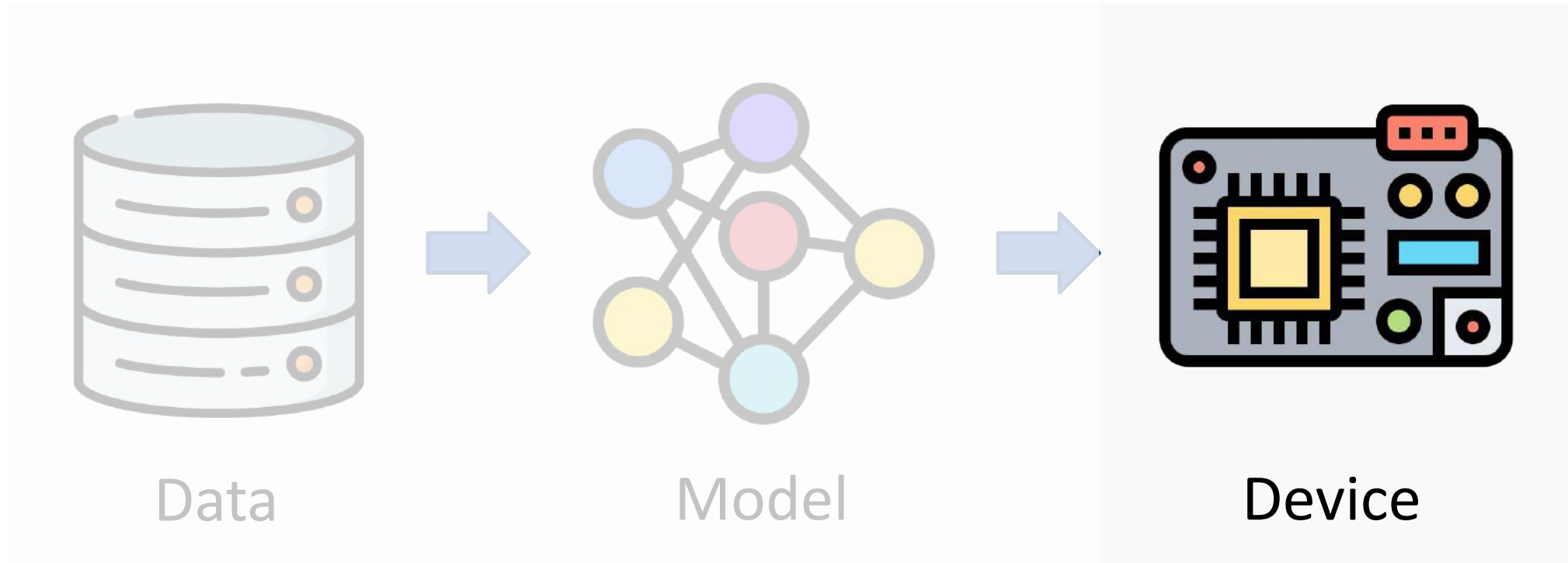
GAN - Generative Adversarial N.

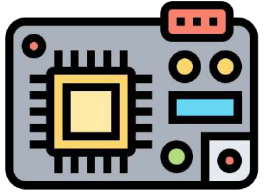


AN - Attention (Transformers)



ML Deployment Pipeline





250 Billion
MCUs today

Hardware (Development Boards)

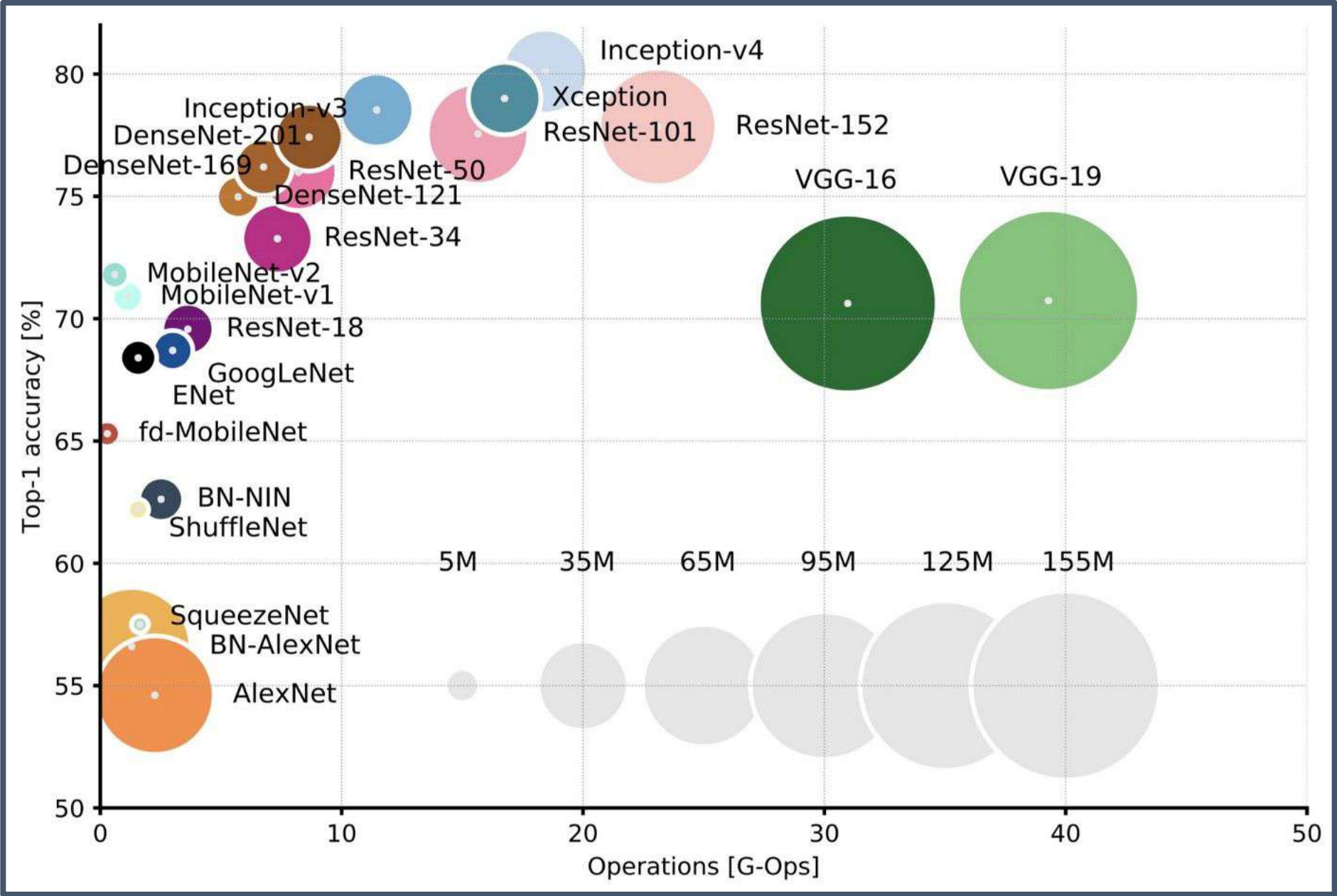


	Raspberry Pico (W)	Arduino Nano Sense	ESP 32	Seeed XIAO Sense / ESP32S3	Arduino Pro
32Bits CPU	Dual-core Arm Cortex-M0+	Arm Cortex-M4F	Xtensa LX6 Dual Core	Arm Cortex-M4F (BLE) Xtensa LX7 Dual Core	Dual Core Arm Cortex M7/M4
CLOCK	133MHz	64MHz	240MHz	64 / 240MHz	480/240MHz
RAM	264KB	256KB	520KB (part available)	256KB / 8MB	1MB
ROM	2MB	1MB	2MB	2MB / 8MB	2MB
Radio	(Yes for W)	BLE	BLE/WiFi	BLE / WiFi (ESP32S3)	BLE/WiFi
Sensors	No	Yes	No	Yes (Sense)	Yes (Nicla)
Bat. Power Manag.	No	No	No	Yes	Yes
Price	\$	\$\$\$	\$	\$\$	\$\$\$\$\$

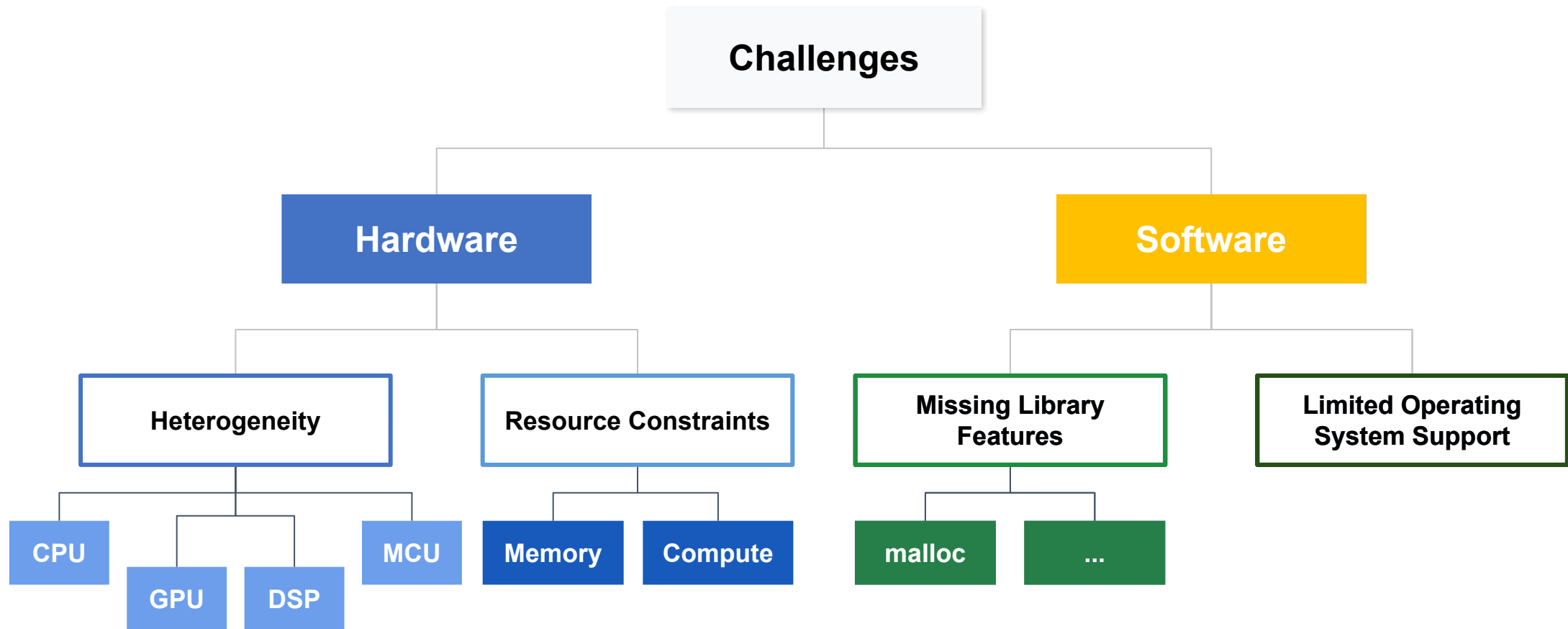
Hardware



	Raspberry Pico (W)	Arduino Nano Sense	ESP 32	Seeed XIAO Sense / ESP32S3	Arduino Pro
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Sensors	No	Yes	No	Yes (Sense)	Yes (Nicla)
Bat. Power Manag.	No	No	No	Yes	Yes
Price	\$	\$\$\$	\$	\$\$	\$\$\$\$\$







Solution

Datasets Preprocessing

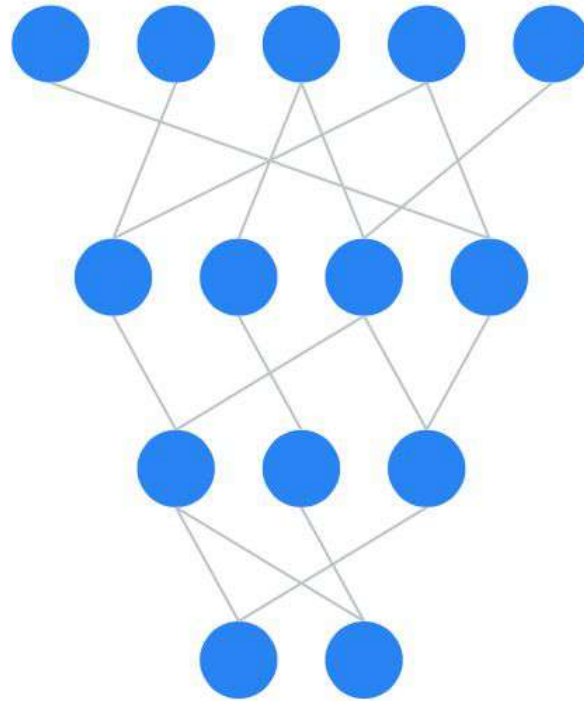
Sound

Vision

Vibration

Text

Quantization Pruning, Distillation

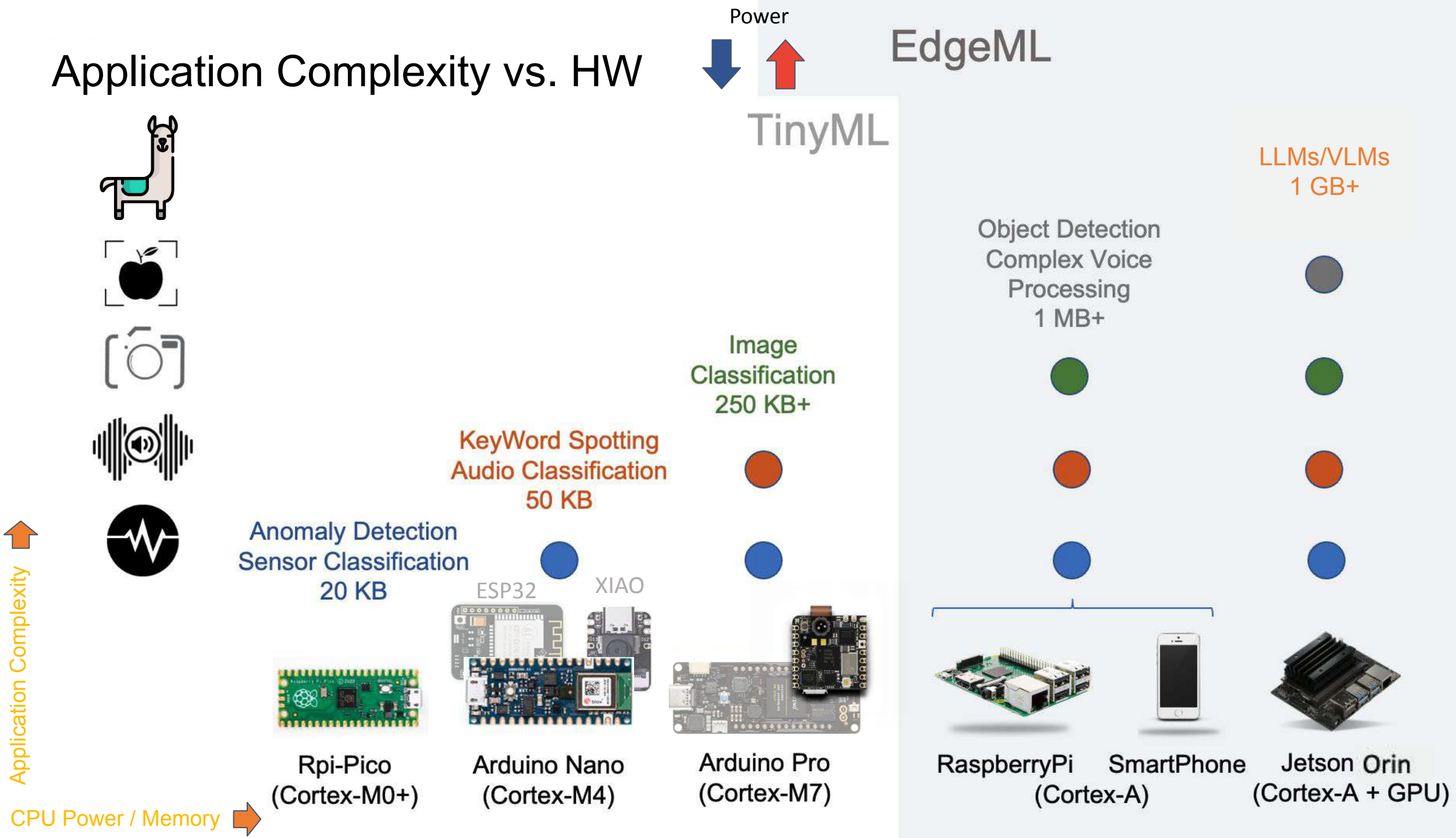


Resource constraints



End-to-end TinyML/EdgeAI Application

Application Complexity vs. HW



Power



EdgeML

TinyML

LLMs/VLMs
1 GB+

1 GB+

Object Detection
Complex Voice
Processing
1 MB+

Complex Voice

Processing

1 MB+

Image
Classification
250 KB+

Classification

250 KB+

KeyWord Spotting
Audio Classification
50 KB

Audio Classification

50 KB

Anomaly Detection
Sensor Classification
20 KB

Sensor Classification

20 KB

FSP32

XIAO

Rpi-Pico
(Cortex-M0+)

(Cortex-M0+)

Arduino Nano (Cortex-M4)

(Cortex-M4)

Arduino Pro (Cortex-M7)

(Cortex-M7)

RaspberryPi (Cortex-M)

(Cortex-A)

SmartPhone

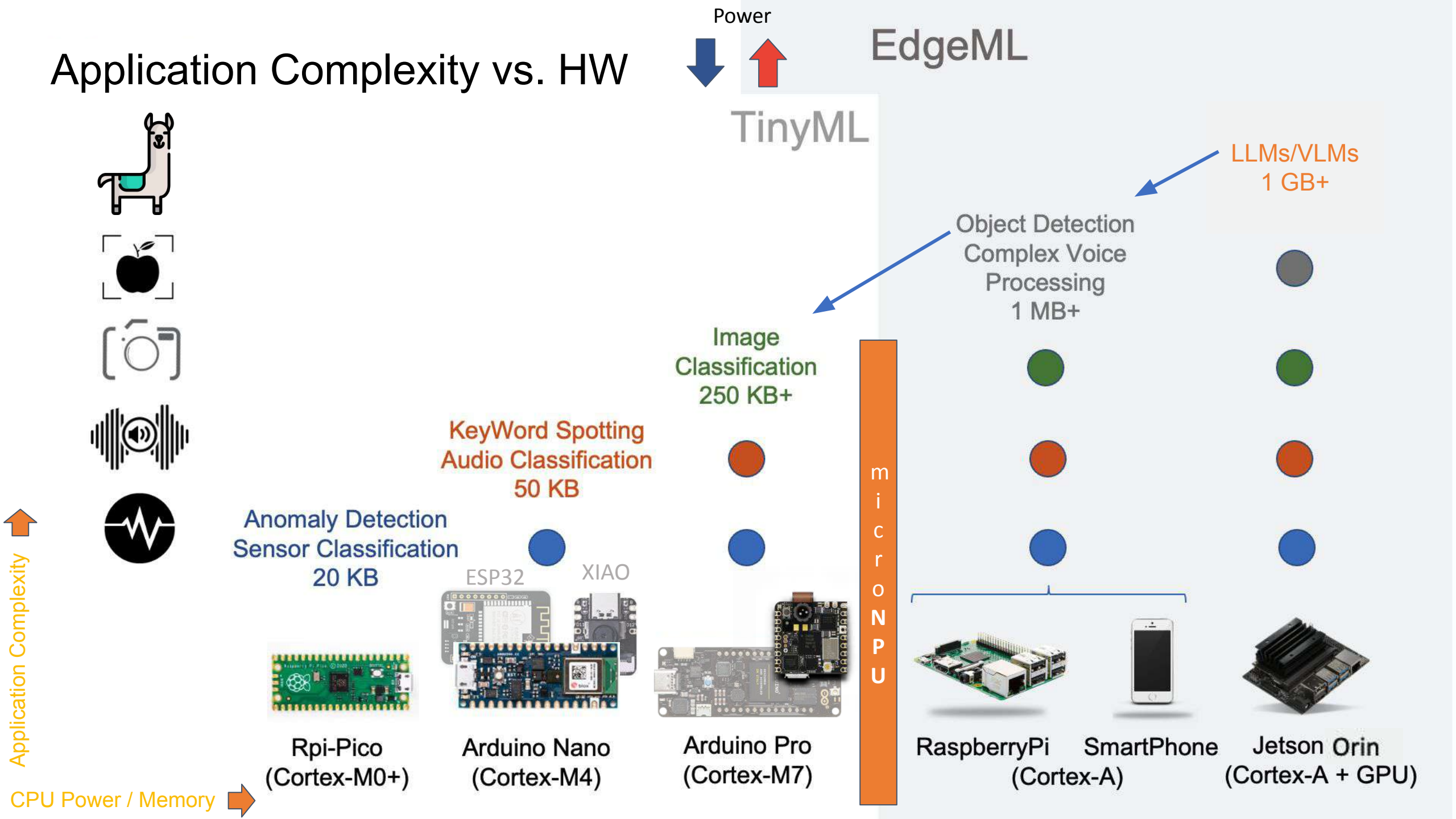
Jetson Orin
(Cortex-A + GPU)

(Cortex-A + GPU)

Application Complexity

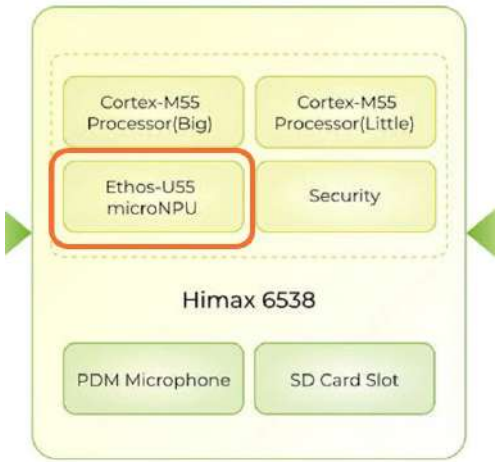
CPU Power / Memory ➡

Application Complexity vs. HW



microNPU

Grove Vision AI v2

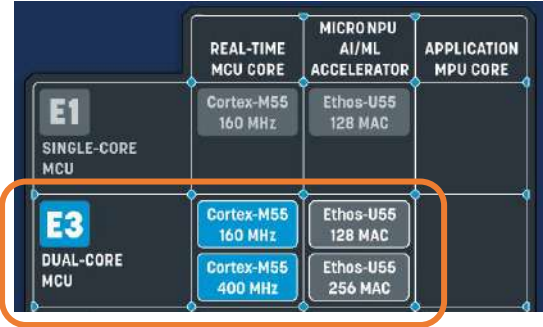


350mW

0.5 TOPS



OpenMV AE3



Alif E3



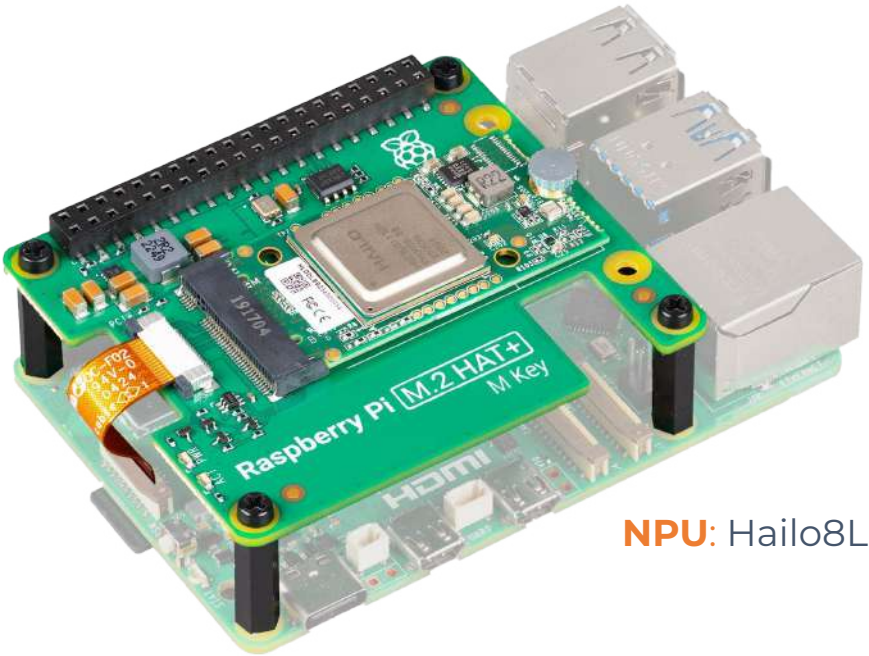
150mW

0.25 TOPS



NN Inference Accelerator

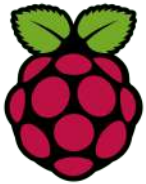
Raspberry Pi AI Kit



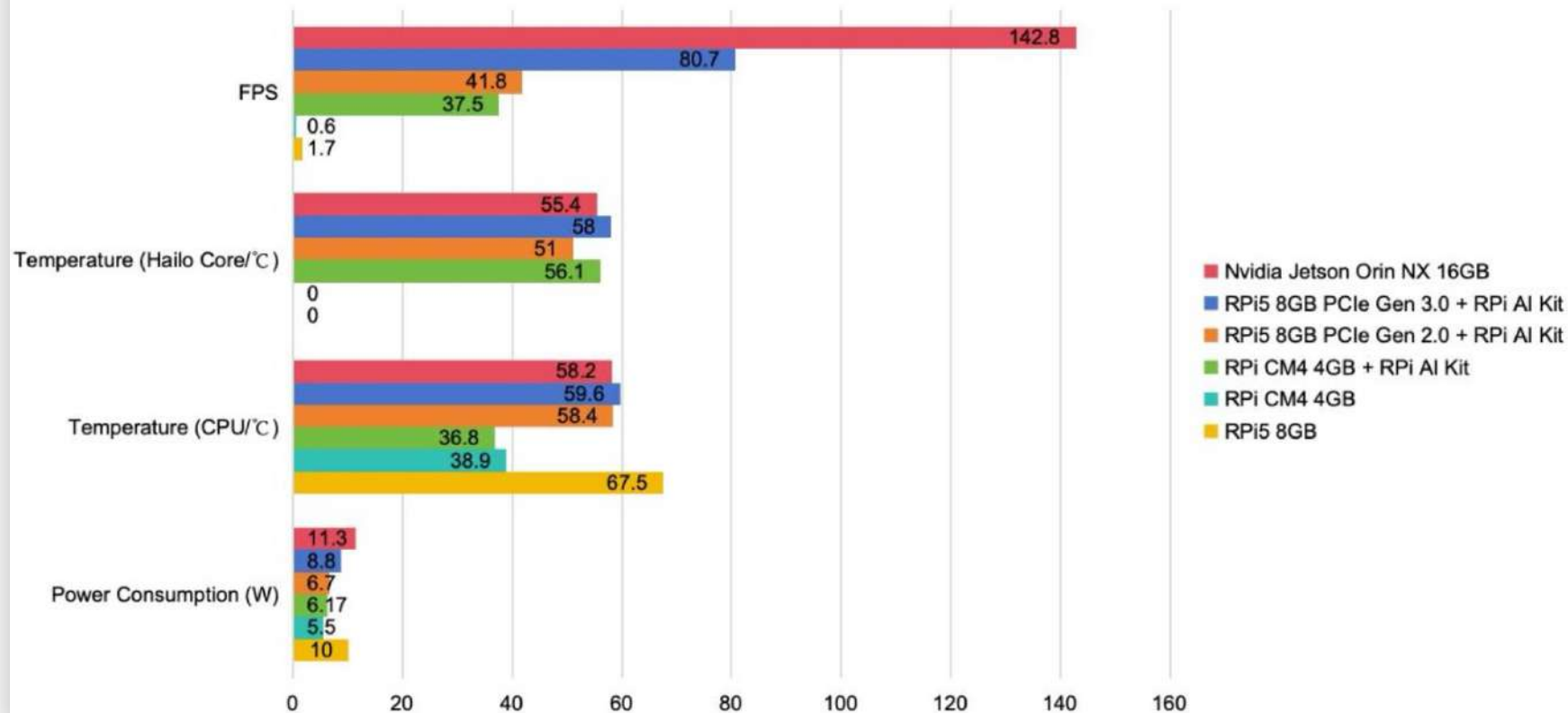
NPU: Hailo8L

30 W

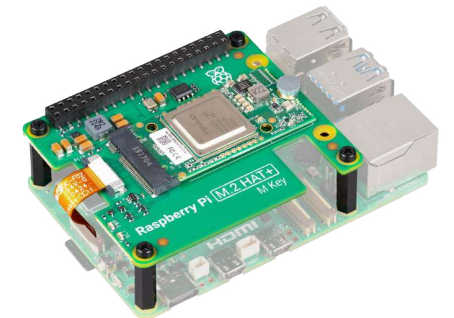
13 - 25 TOPS



Object Detection Benchmark on Raspberry Pi and Nvidia Jetson Orin NX with YOLOv8s



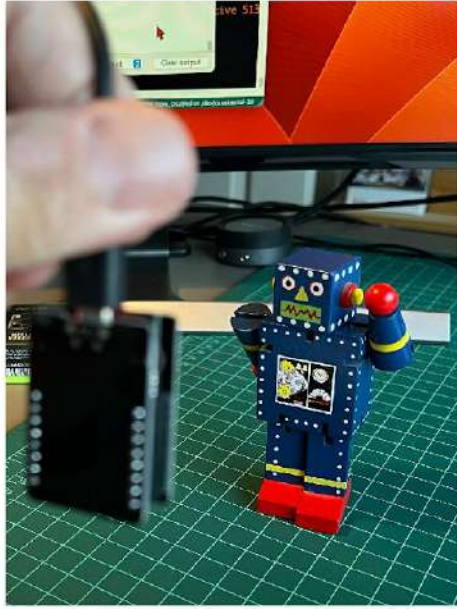
GPU: NVIDIA Ampere
100 TOPS



NPU: Hailo8L*

* **Note:** Hailo-10H for LLMs was launched in 2025

Tiny Image Classification Benchmark (MobileNetV2 96x96 0.1)



Classification: 687 ms

1.5 FPS



ESP - CAM
Xtensa LX6
240 MHz

300 mW



Classification: 142 ms

7.0 FPS



XIAO ESP32S3
Xtensa LX7
240 MHz

525 mW



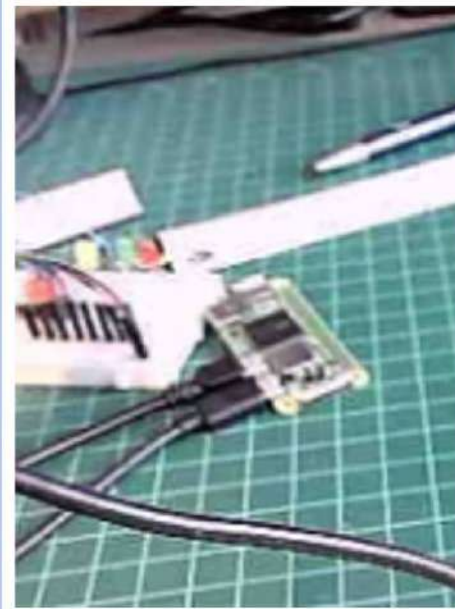
Classification: 86 ms

11.6 FPS



Nicla-Vision
ARM M7
480 MHz

600 mW



Classification: 11.0 ms

91.0 FPS



Raspi Zero W2
ARM A53
1 GHz

1,500 mW



Classification: 6 ms

167 FPS



Grove Vision AI V2
ARM Ethus-U55
400 MHz

420 mW

Real-World Applications

Examples

[illegible]

Students & researchers solving local problems with global tools

Real-World Applications

Agriculture

Healthcare

Industry

Environment

Real-World Applications

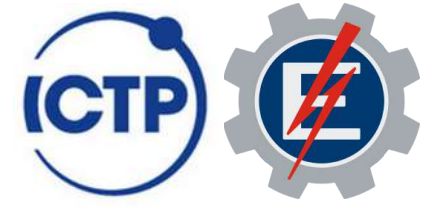
Agriculture

Healthcare

Industry

Environment

Fighting Dengue and Malaria



Challenge

Mosquito-borne diseases affect 700M+ people/year

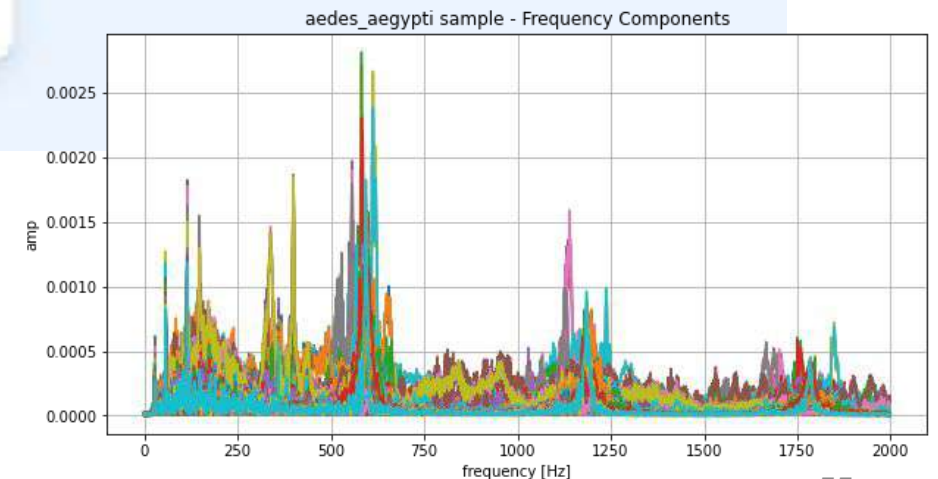
Solution

Audio classification of mosquito wingbeats

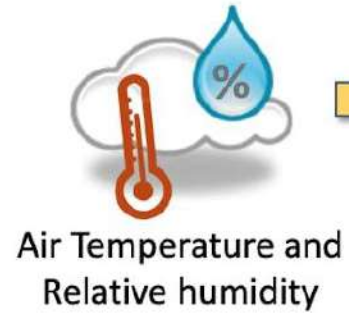
"TinyML where internet doesn't reach"

Impact

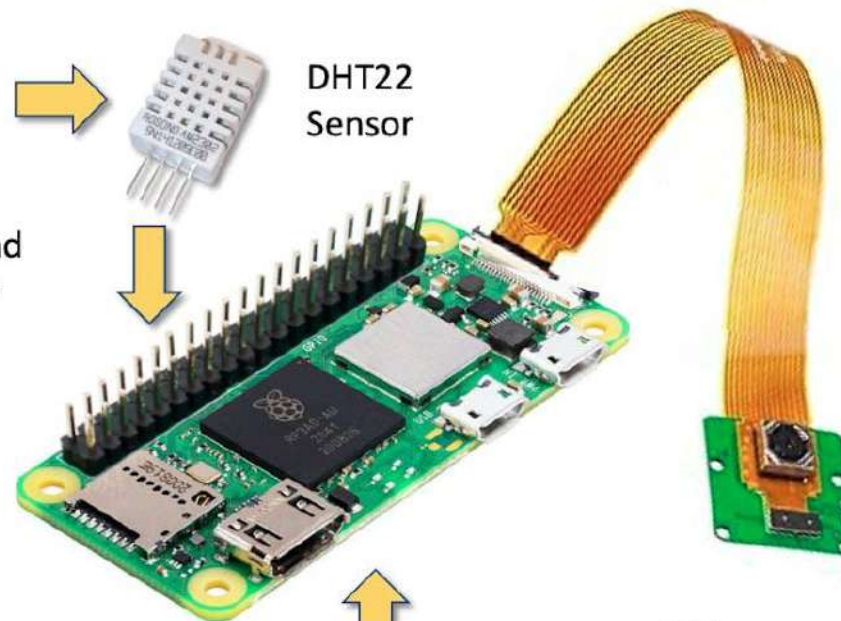
- Species identification
- Early warning system
- Works in the field
- Cited by Science Magazine



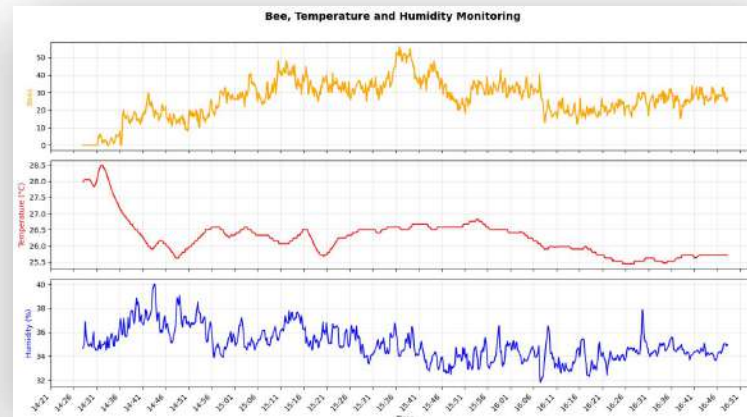
Bee Counting



Air Temperature and
Relative humidity



DHT22
Sensor



sampleFreq → 10 s

José Anderson Reis
UNIFEI Master's Student

Guilherme Fernandes
UNIFEI Grad Student

Coffee Disease Classification



João Vitor Yukio Bordin Yamashita
Engenheiro - UNIFEI

<https://www.hackster.io/Yukio/coffee-disease-classification-with-ml-b0a3fc>

LSTM Phenolic Sponge Moisture



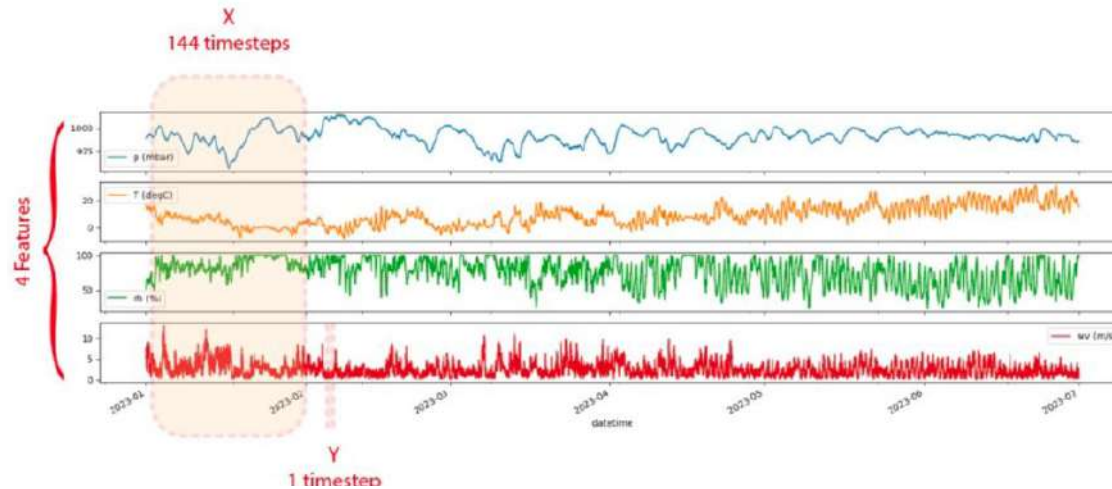
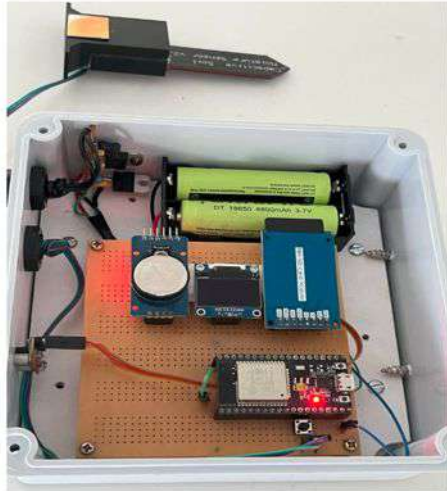
FURG



UTEC



UNRaf



Gustavo Pablo Castro Abdallah
PRIA

Ant Detection



FURG



UTEC



UNRaf



Juan Camilo Abedala
PRIA

Real-World Applications

Agriculture

Healthcare

Industry

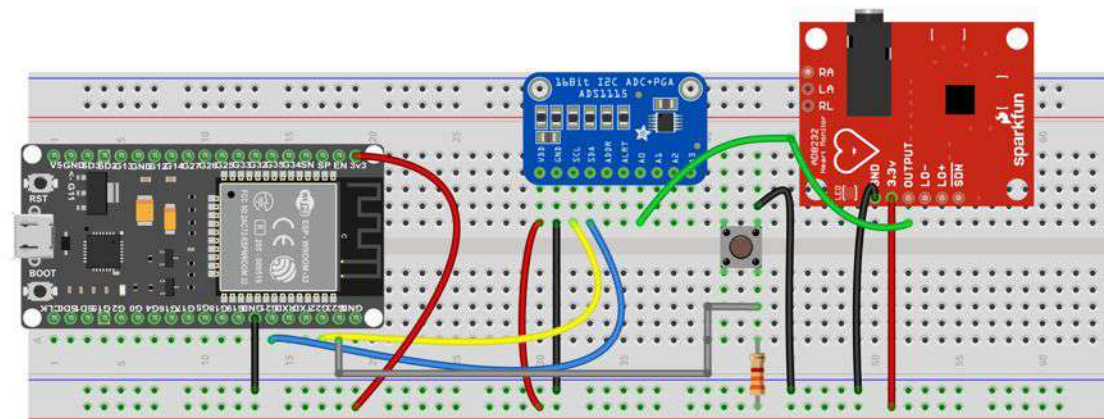
Environment



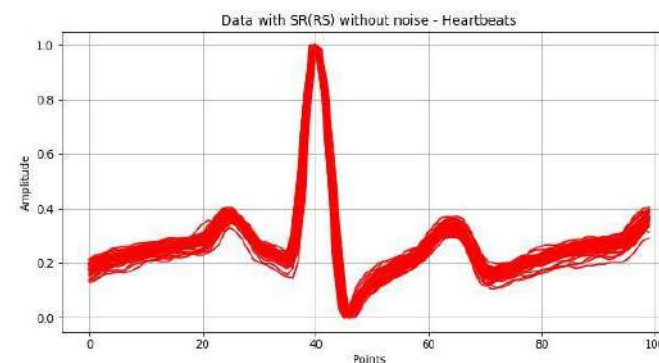
Single Lead Heart Rate Monitor



[Atrial Fibrillation Detection on ECG using TinyML](#)
Silva et al. UNIFEI 2021

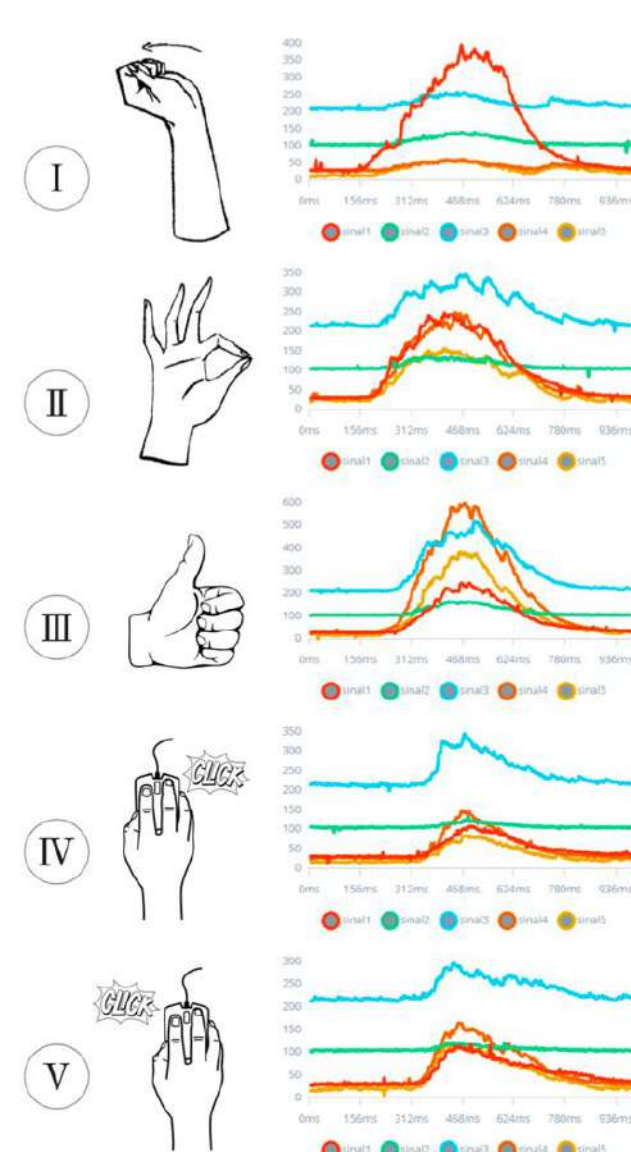
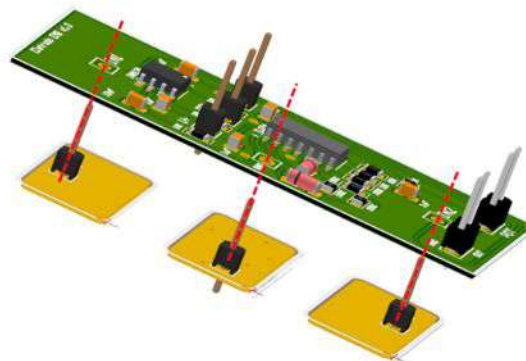
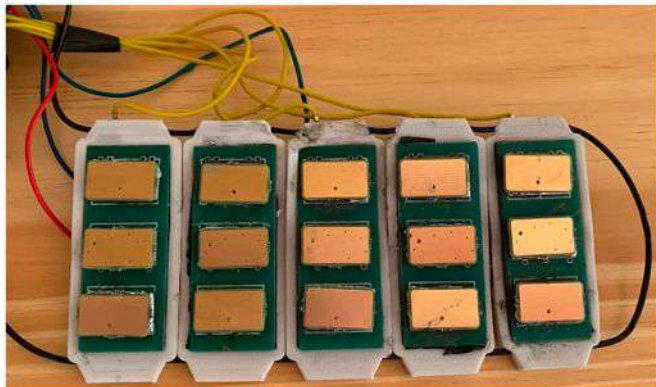
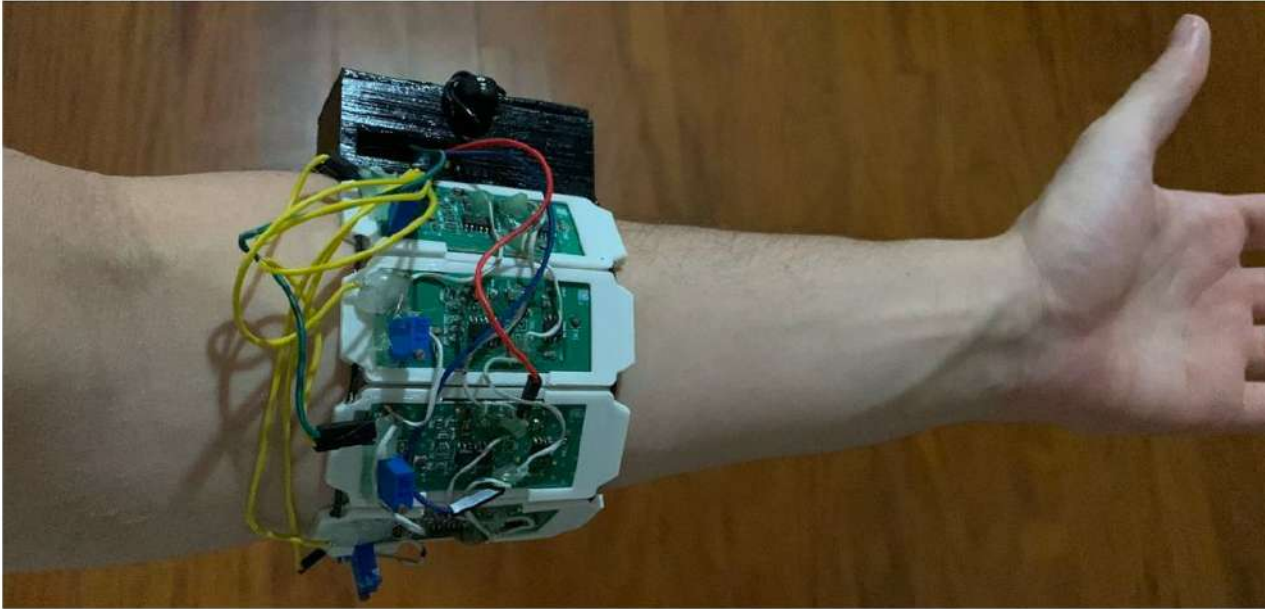


fritzing



Guilherme Silva
Matheus Lima
Engenheiros - UNIFEI

Surface electromyography



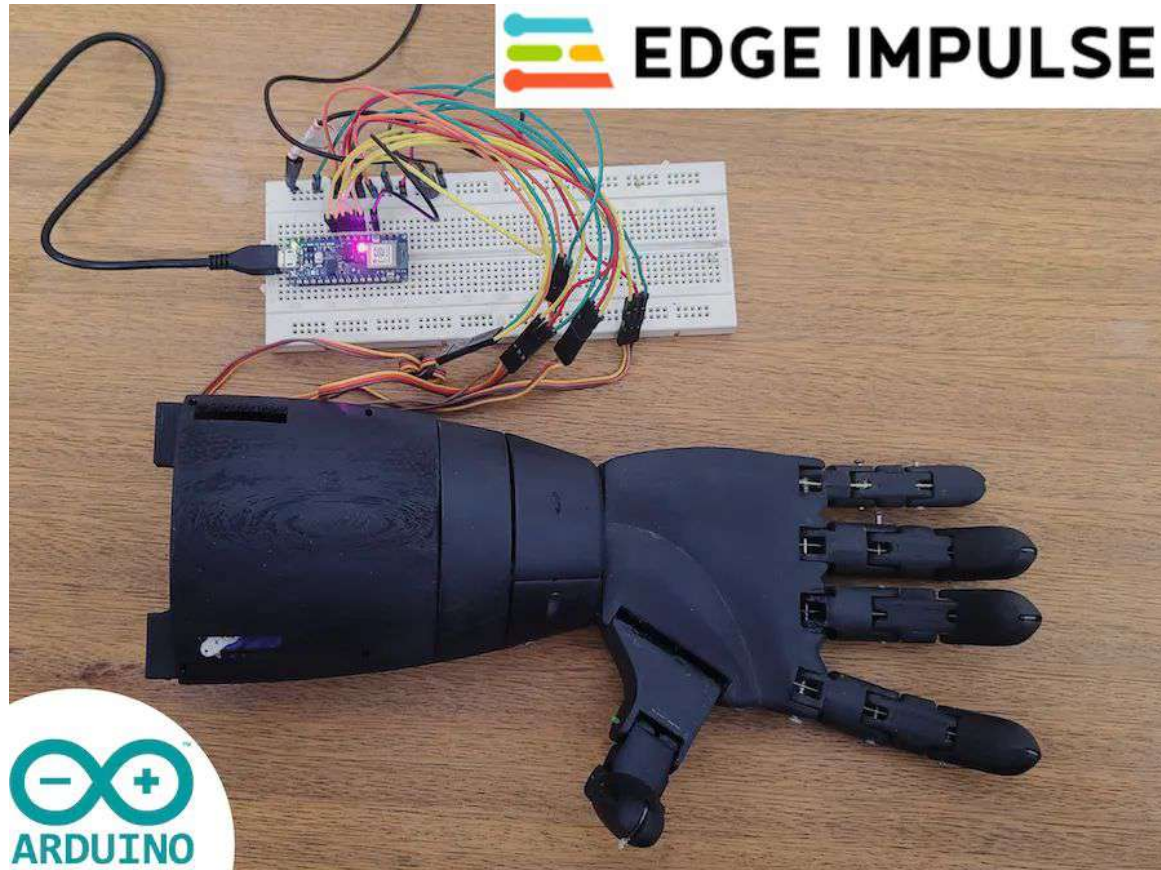
Mateus F. Delangélica e
Renato M. Neto, UNIFEI

Personal Trainer



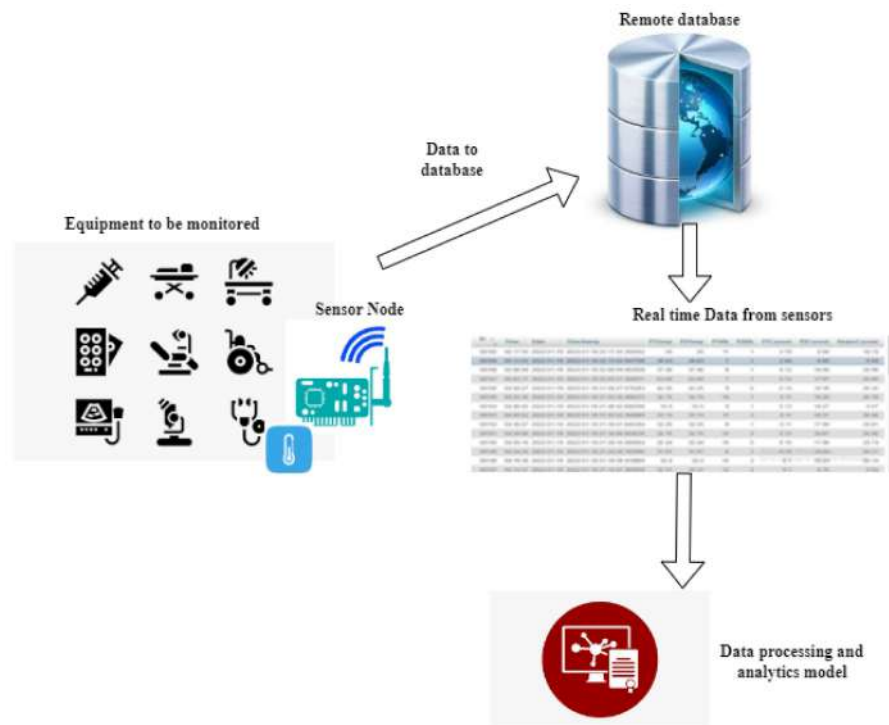
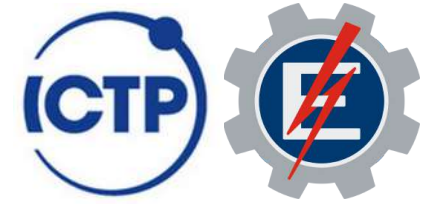
Ricardo Magno C. Junior
Luiz Fernando Kikuchi
UNIFEI

Bionic Hand Voice Commands



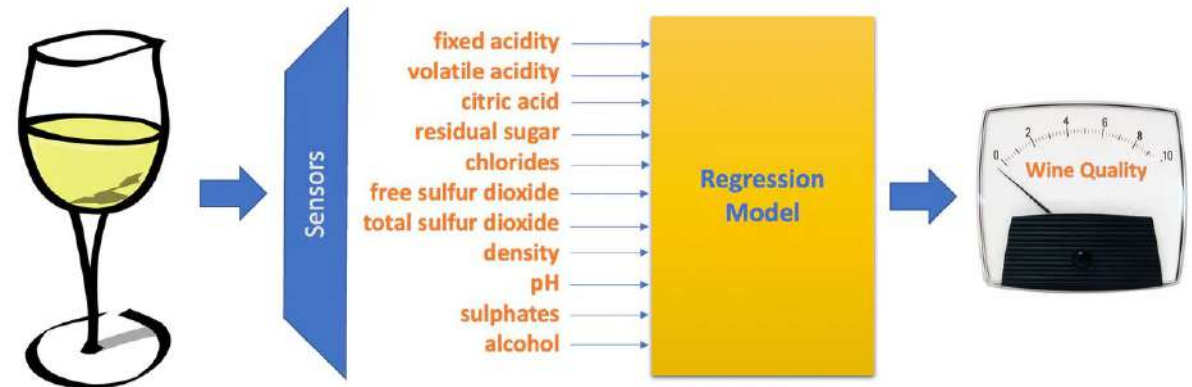
<https://www.hackster.io/ex-machina/bionic-hand-voice-commands-module-w-edge-impulse-arduino-aa97e3>

Regression on TinyML



[On-Device IoT-Based Predictive Maintenance Analytics Model: Comparing TinyLSTM and TinyModel from Edge Impulse](#)

Sensor fusion



[TinyML Made Easy: Exploring Regression - White Wine Quality](#)

Real-World Applications

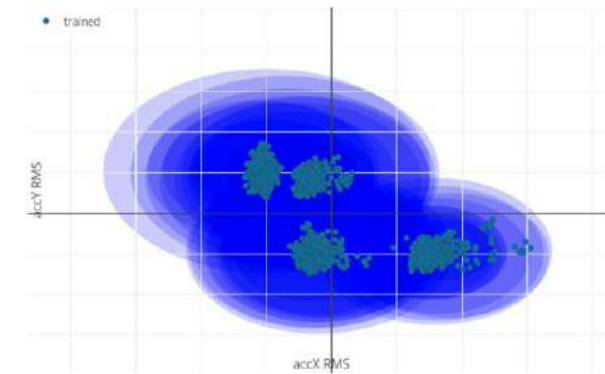
Agriculture

Healthcare

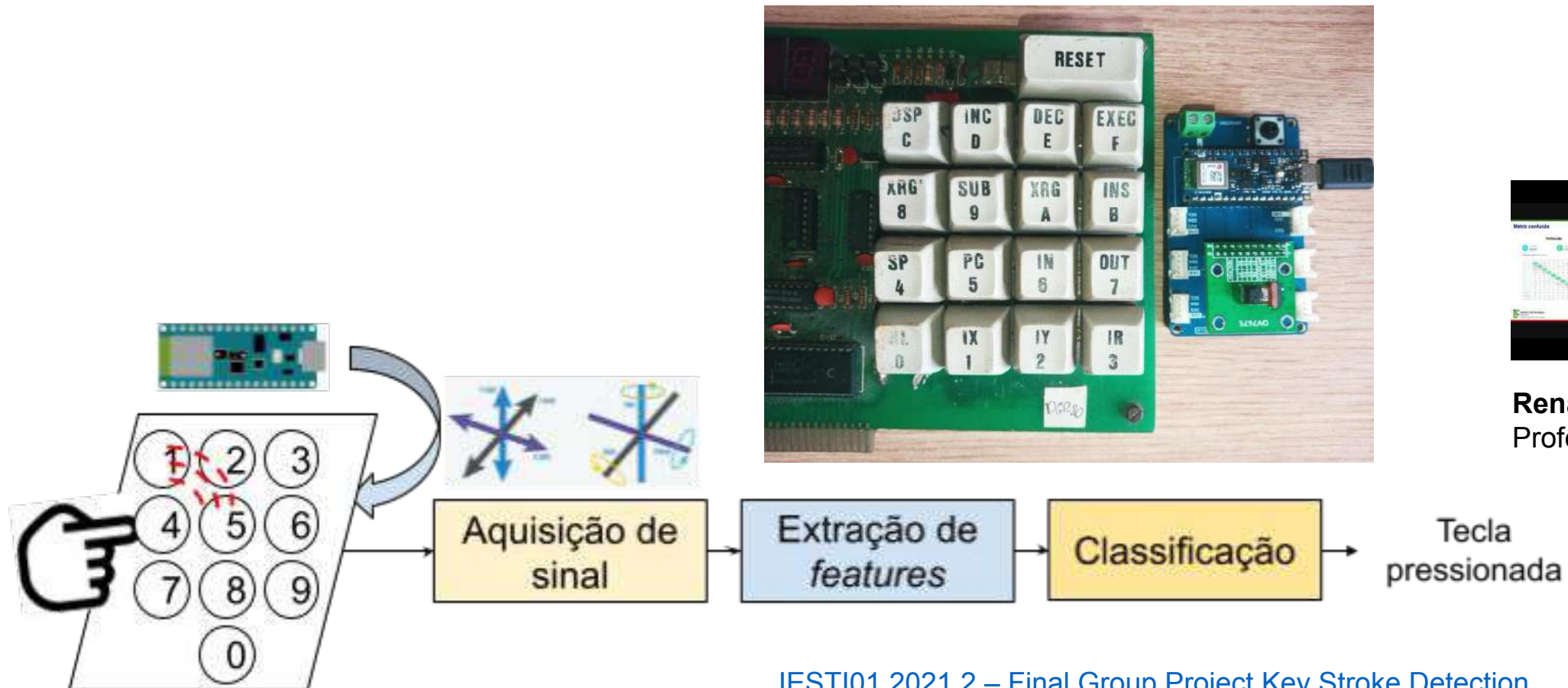
Industry

Environment

Industrial – Anomaly Detection



Keystroke **Sound** Detection



Renam Castro
Professor IFESP



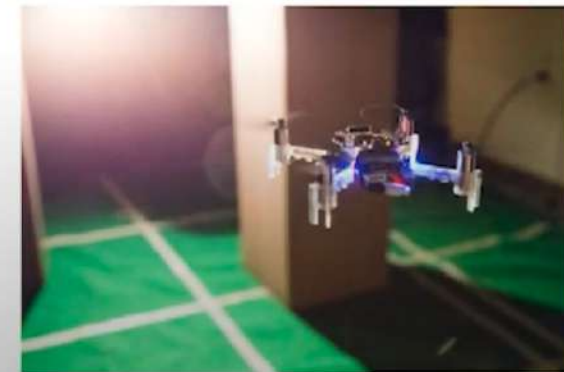
Reinforcement on TinyML

Deep Reinforcement Learning for Autonomous Source Seeking on a Nano Drone

Bardienus P. Duisterhof^{1,3} Srivatsan Krishnan¹ Jonathan J. Cruz¹ Colby R. Banbury¹ William Fu¹

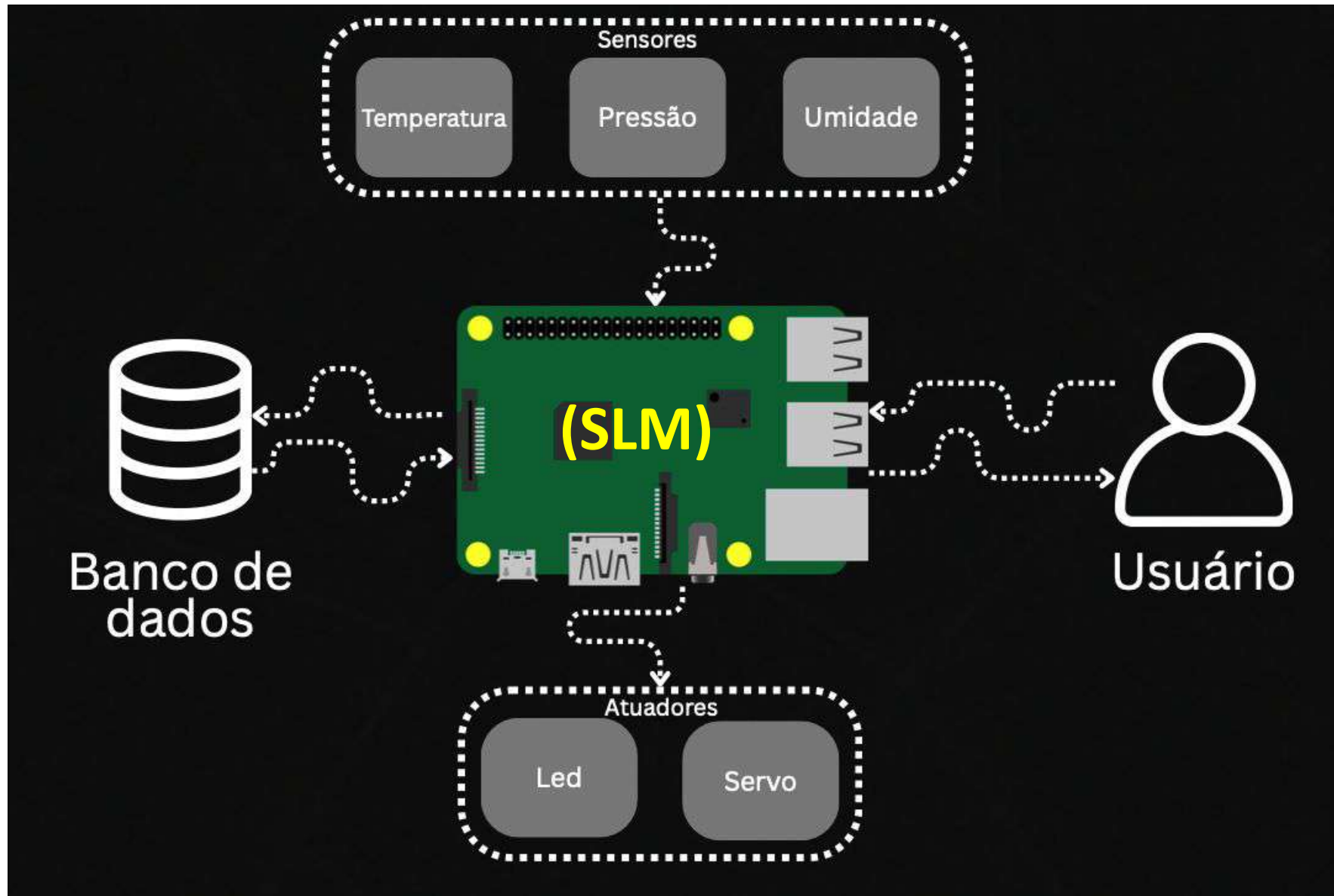
Aleksandra Faust² Guido C. H. E. de Croon³ Vijay Janapa Reddi^{1,4}

¹Harvard University, ²Robotics at Google, ³Delft University of Technology, ⁴The University of Texas at Austin



<https://arxiv.org/abs/1909.11236>

<https://youtu.be/wmVKbX7MOnU>



João Pedro Fonseca
UNIFEI Grad Student (IESTI)

Matheus Souza
UNIFEI Grad Student (IMC)

Real-World Applications

Agriculture

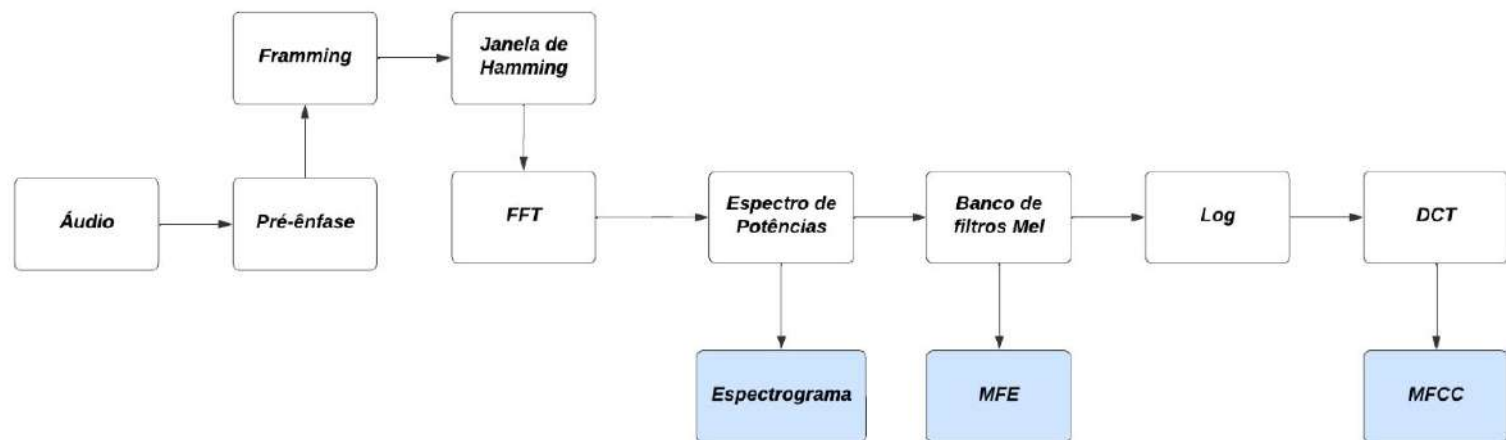
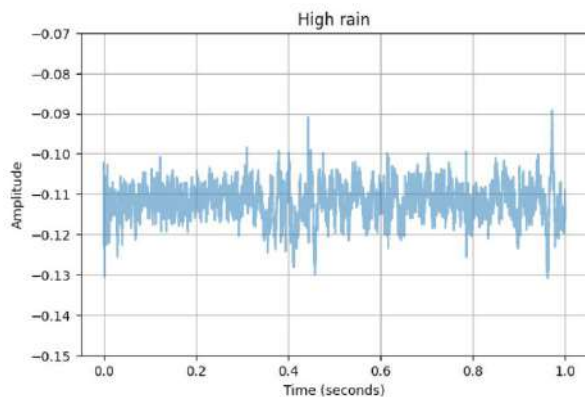
Healthcare

Industry

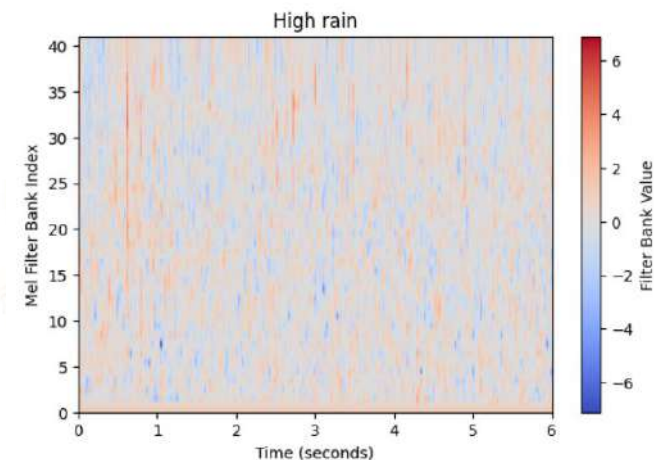
Environment



Measure rainfall using sound detection



```
Edge Impulse standalone inferencing
run_classifier returned: 0
Timing: DSP 630 ms, inference 47 ms,
Predictions:
  HighRain: 0.99609
  LowRain: 0.00000
  MediumRain: 0.00000
  NoRain: 0.00000
```



The Future of the EdgeAI

The Newest Frontier: GenAI at the Edge

Desktop Reference

PC Linux (i7): 20 tokens/s

Mac (M1-Pro): 111 tokens/s

Running Llama 3.2:1B

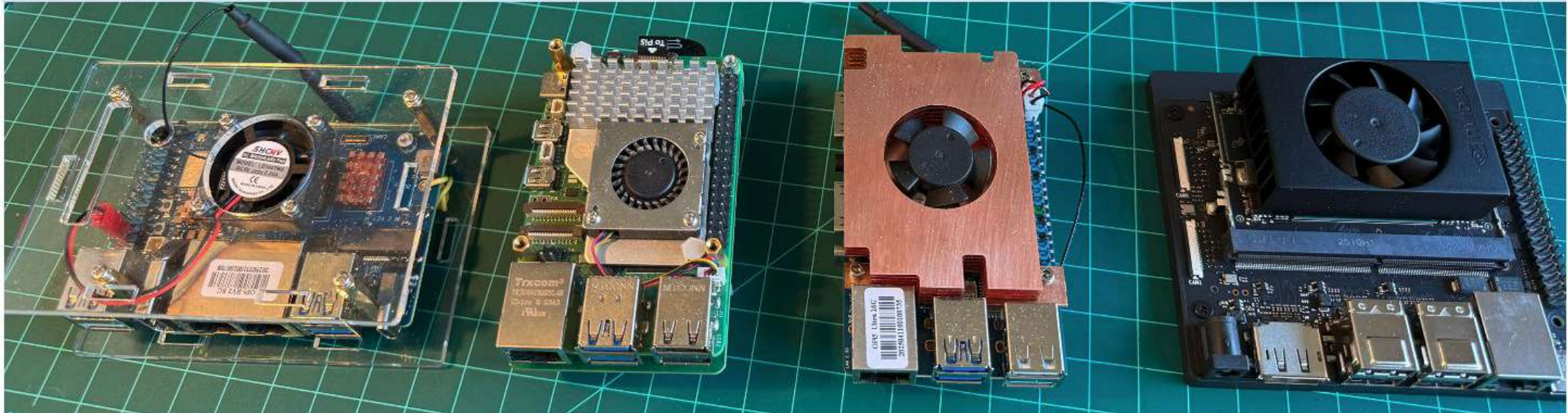
(*) Running on GPU

Orange Pi RV2
1 token/s

Raspi-5
7.5 tokens/s

Orange Pi 5 Ultra
12 tokens/s

Jetson Orin Nano
26 tokens/s (*)



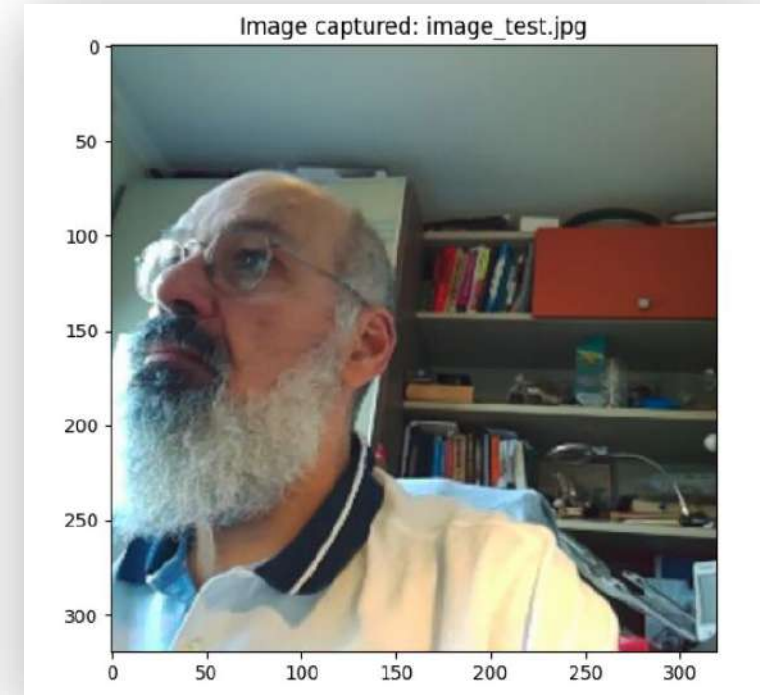
Intelligent conversations on maker hardware

See It In Action: Vision Description

```
marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.local — 87x18
(ollama) mjrovai@raspi-5-SD:~ $ ollama run moondream --verbose
>>> describe the image: /home/mjrovai/Documents/OLLAMA/image_test.jpg"
Added image '/home/mjrovai/Documents/OLLAMA/image_test.jpg'

The image features a man with glasses and a white beard, who is wearing a
collared shirt. He appears to be looking off into the distance or at something
out of frame. The room has several bookshelves filled with numerous books,
indicating that he might have an interest in reading or studying.

total duration:      54.36227837s
load duration:       63.257535ms
prompt eval count:    743 token(s)
prompt eval duration: 48.361595541s
prompt eval rate:     15.36 tokens/s
eval count:          62 token(s)
eval duration:        5.935988132s
eval rate:            10.44 tokens/s
>>> Send a message (/? for help)
```



Moondream 2 on Raspberry Pi

- Describes images in natural language
- Answers questions about photos
- Runs completely offline
- 10 tokens/second

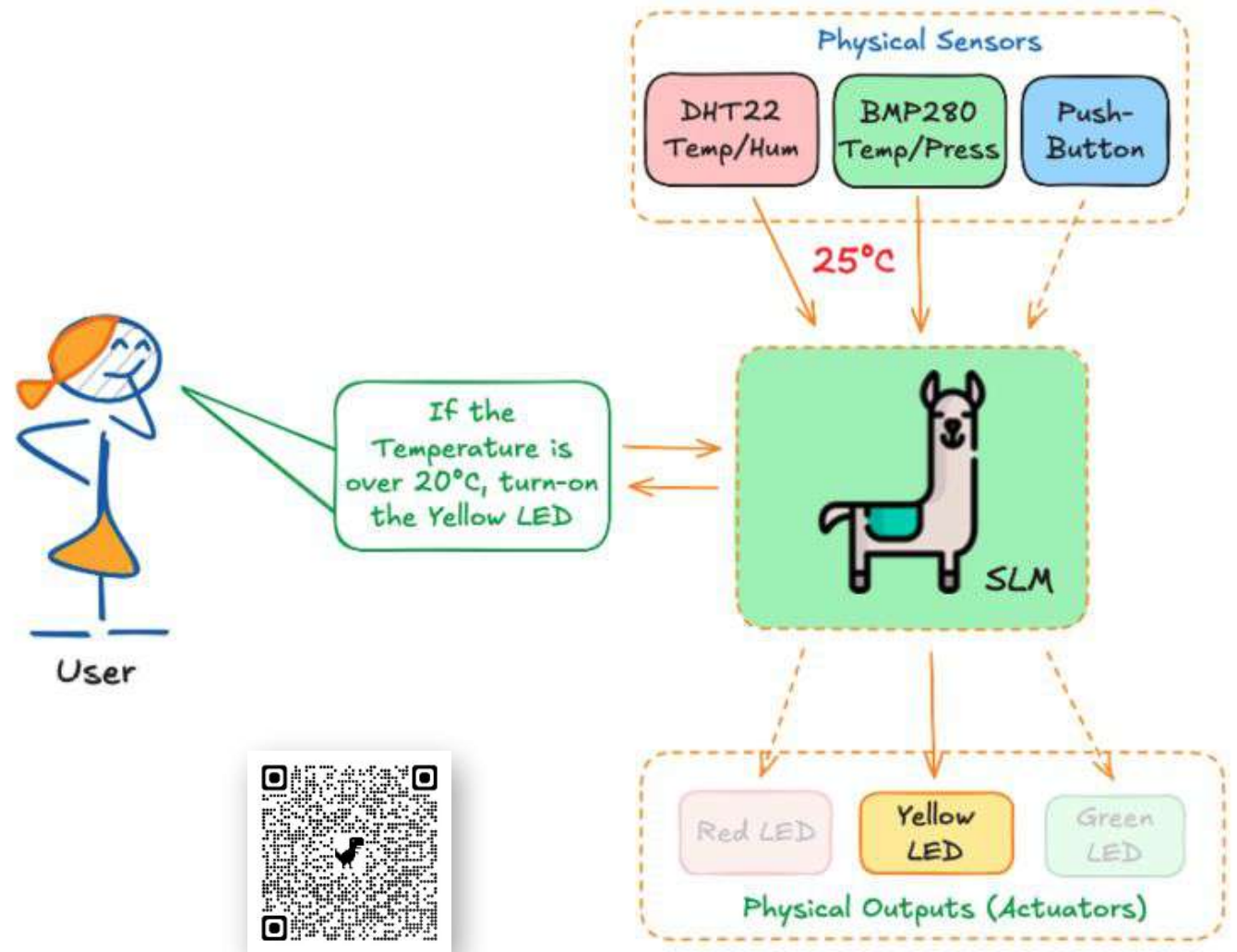
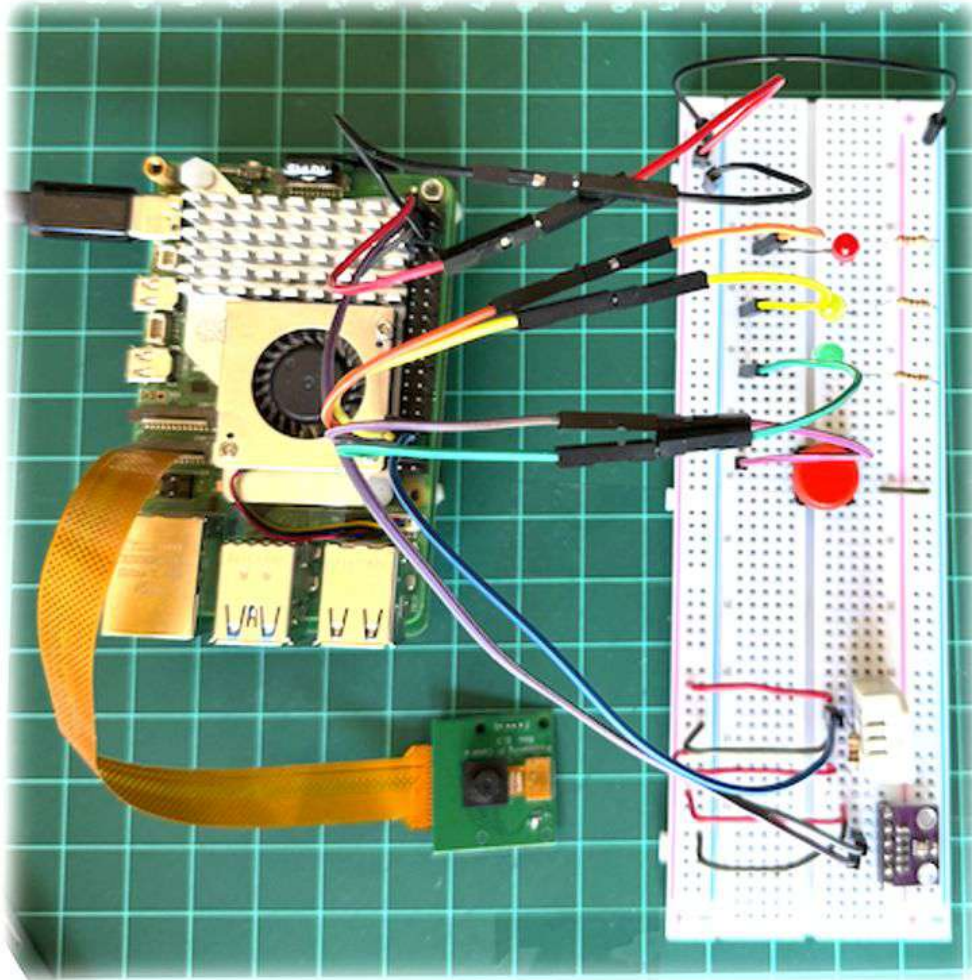
```
marcelo_rovai — mjrovai@raspi-5-SD: ~ — ssh mjrovai@raspi-5-SD.loca...
>>> what is the color of the shirt?

The man's shirt is white.

total duration:      1m15.308297242s
```



Physical Computing Meets Intelligence

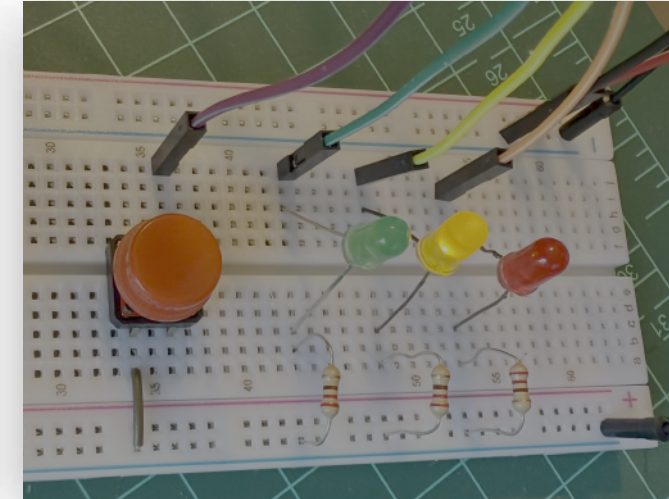


Sensors → AI → Actions

```
marcelo_rovai — mjrovai@raspi-5: ~/Documents/Smart_iot/Basic — ssh mjrovai@192.168.4.209 — 97x17

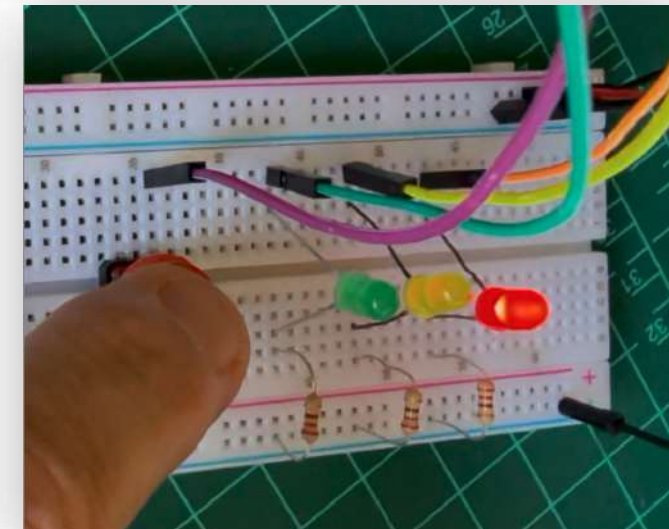
=====
Command: if the temperature is above 20°C, turn on yellow led
=====
Time: 10:25:19
DHT22: 23.2°C, 42.4%
BMP280: 23.7°C, 905.2hPa
Button: not pressed
SLM Response: To turn on the yellow LED, as the temperature (23.7°C) is above 20°C, I will proceed with activating it.

Activate Yellow LED
LED Status: R=off, Y=ON, G=off
=====
Command: 
```



```
marcelo_rovai — mjrovai@raspi-5: ~/Documents/Smart_iot/Basic — ssh mjrovai@192.168.4.209 — 97x18

=====
Command: if the button is pressed, turn on red led
=====
Time: 10:04:01
DHT22: 22.8°C, 43.3%
BMP280: 23.4°C, 905.2hPa
Button: pressed
SLM Response: 1. You want the system to turn on the red LED when the button is pressed.
2. No request about sensor data.
3. Activate Red LED
LED Status: R=ON, Y=off, G=off
=====
Command: 
```



Arduino Q's

UNO-Q



VENTUNO-Q

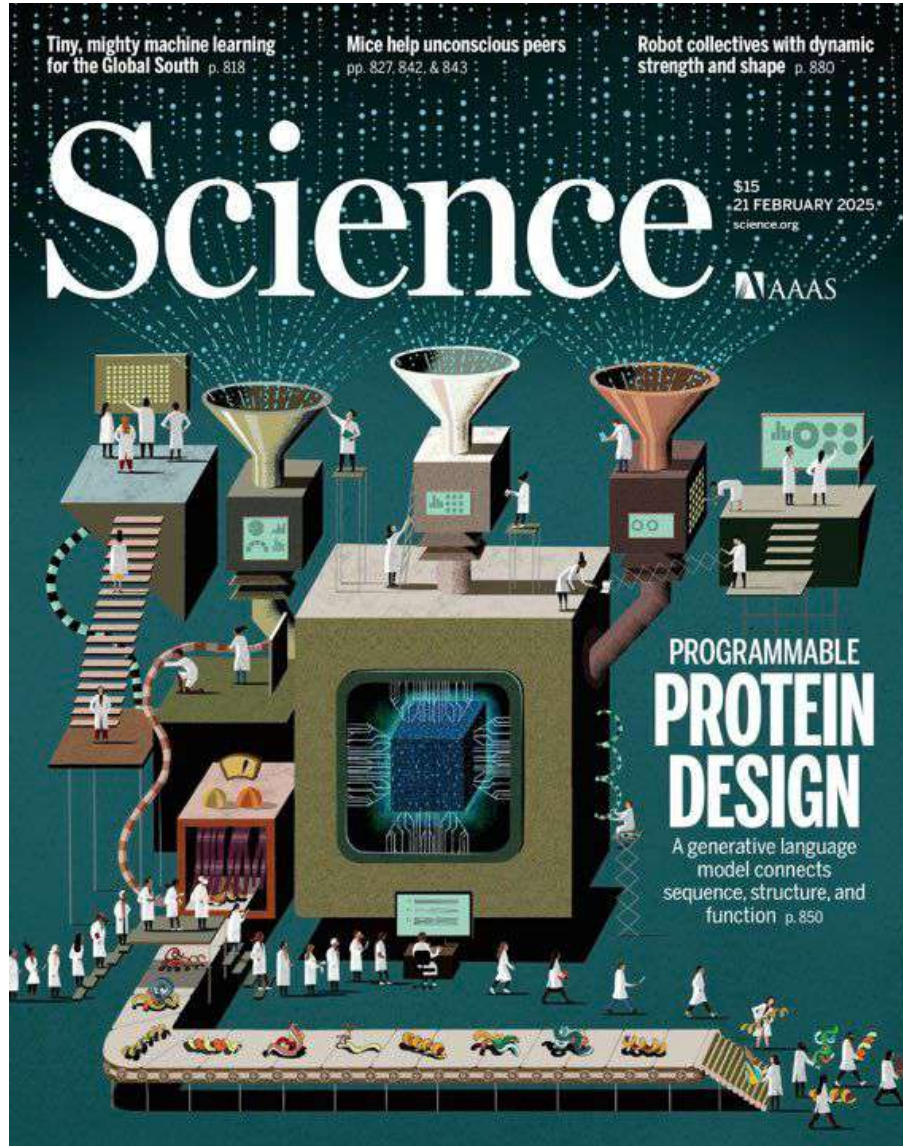


Qualcomm
MPU



STMicroelectronics
MCU





<https://www.science.org/content/article/what-s-tinyml-global-south-s-alternative-power-hungry-pricey-ai>

CUTTING AI DOWN TO SIZE



A \$14 chip
incorporating tinyML
AI models.
actual size shown.



The Future of AI is in Our Hands

Not in massive data centers, but

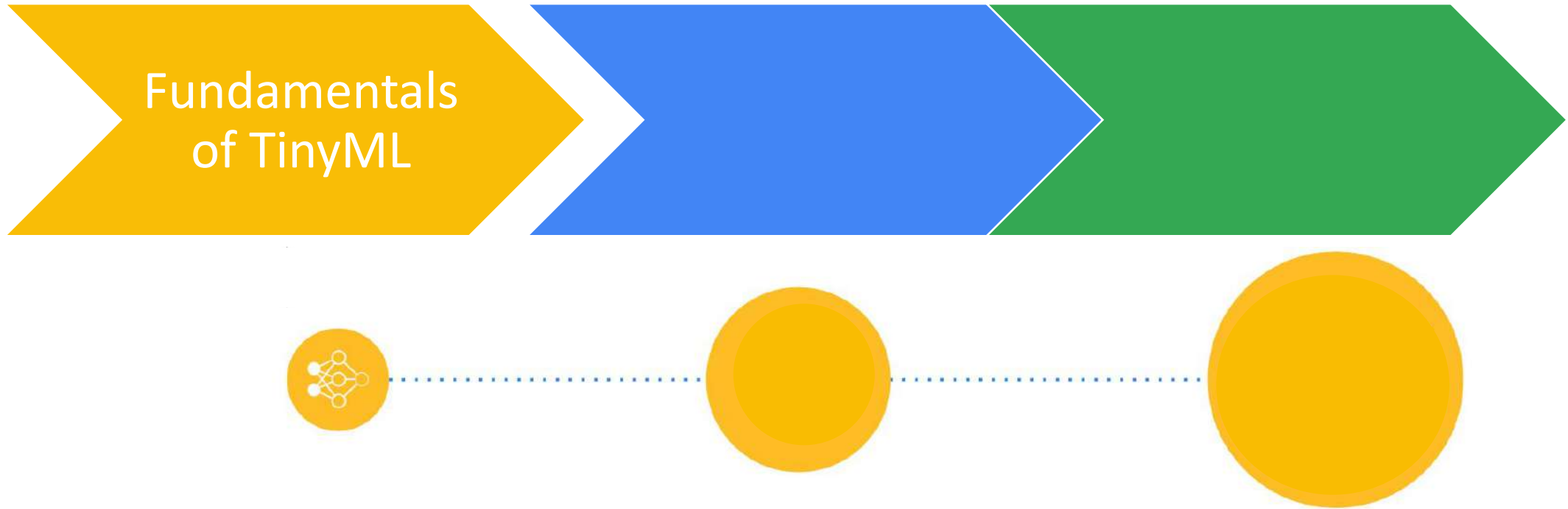
- ✓ In classrooms across Asia
- ✓ On farms in Africa
- ✓ In clinics in rural Latin America
- ✓ In the hands of students like you

Start building today. The tools are free.
The impact is limitless.

What will We learn?

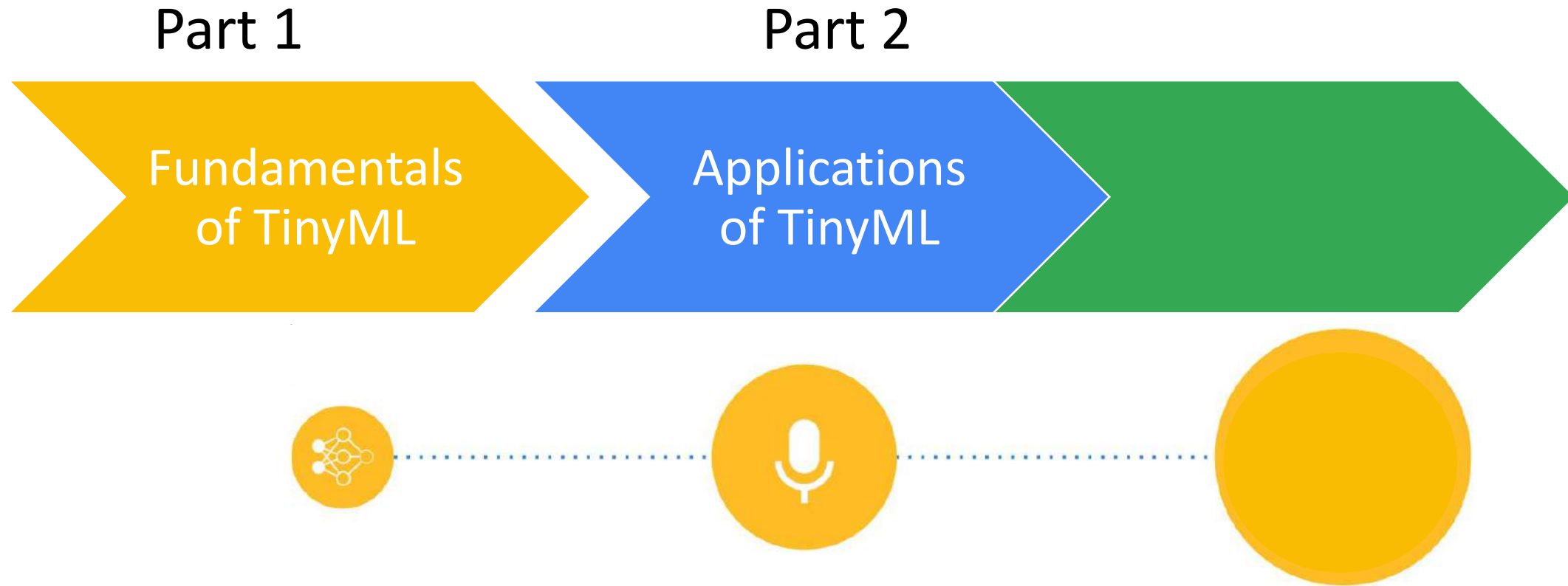
What will We learn?

Part 1



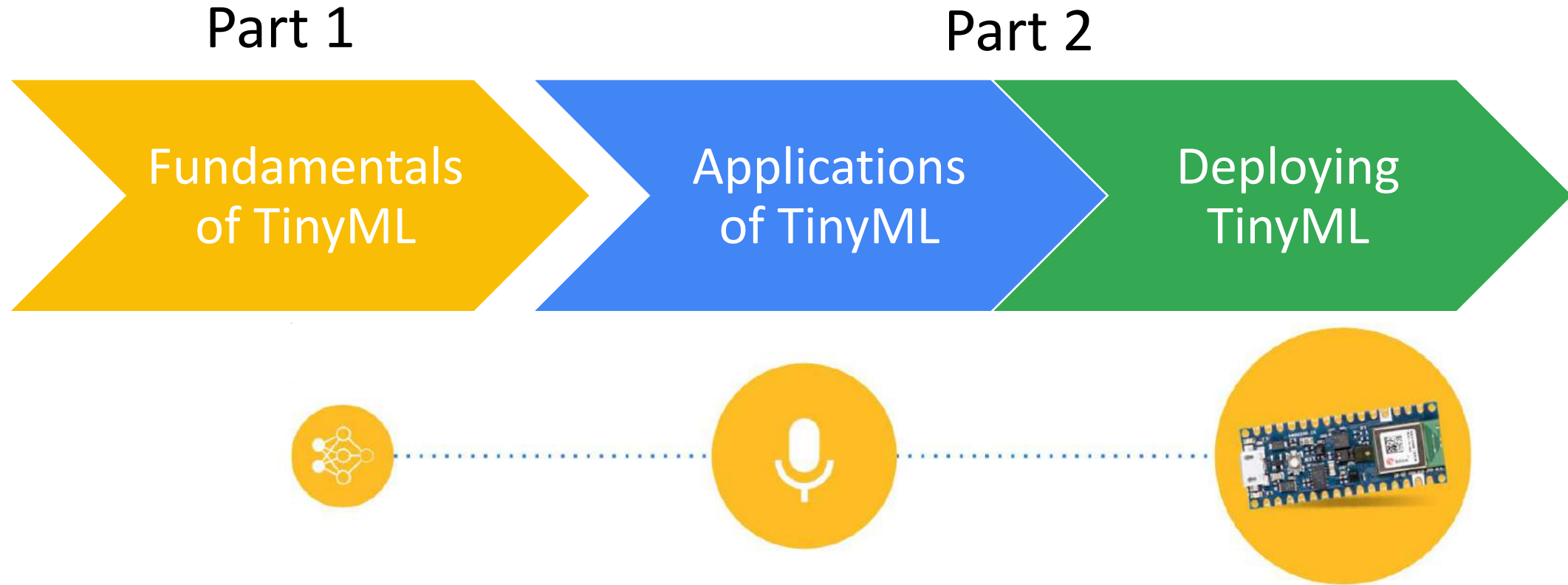
Part 1 is all about talking about what is the language of machine learning

What will we learn?



In Part 2, we will get a sneak peek into the variety of different TinyML applications, as keyword spotting (“Alexa”), gesture recognition, understand how to leverage the sensors, and so forth.

What will we learn?



In Part 2, we will **also** learn how to deploy models on a real microcontroller. Along the way we will explore the challenges unique to and amplified by TinyML (e.g., preprocessing, post-processing, dealing with resource constraints).

Background Requirements

Part 1

Fundamentals of TinyML

- Python
- TensorFlow
- Google Colab
 - Jupyter Notebook

Part 2

Applications of TinyML

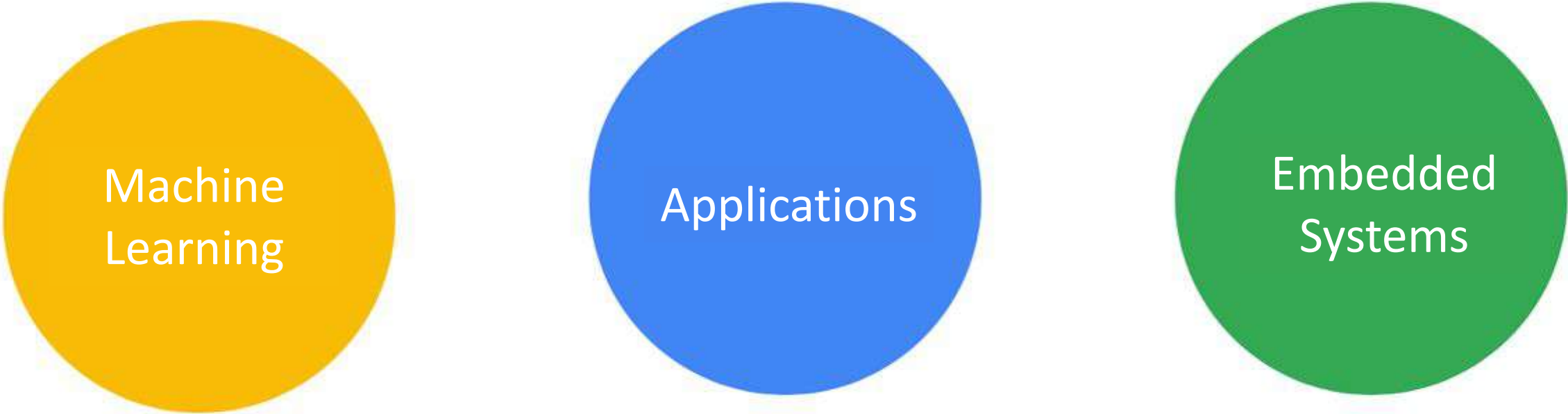
- Python
- TensorFlow (LiteRT)
- Google Colab
- Edge Impulse Studio
- Seeed SenseCraft AI

Deploying TinyML

- Python
- TensorFlow (LiteRT for Microcontrollers)
- Google Colab
- Edge Impulse Studio
- Seeed SenseCraft AI
- C/C++ => IDE (as Arduino)

This course combines **computer science** with **engineering** to feature real-world application case studies that examine the challenges facing **TinyML deployments**.

What areas we will learn?



Machine
Learning

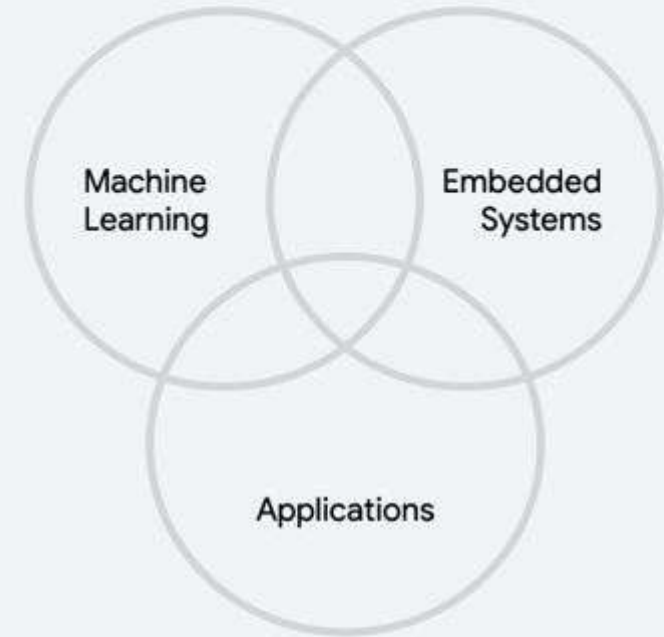
Applications

Embedded
Systems

We will learn the **fundamentals of each of these areas**, just enough to focus on the goal of being able to build **TinyML applications**.

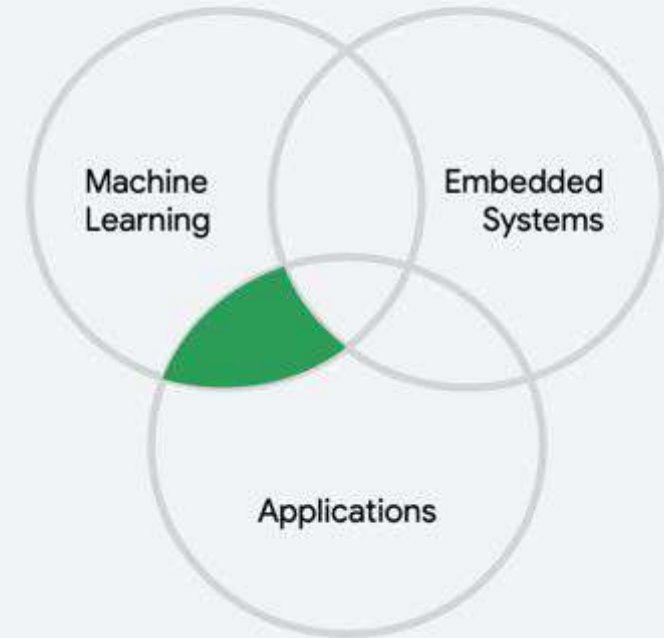
Interactions

In addition, we will bring these diverse topics together to reveal the interesting learnings at the various **intersections**



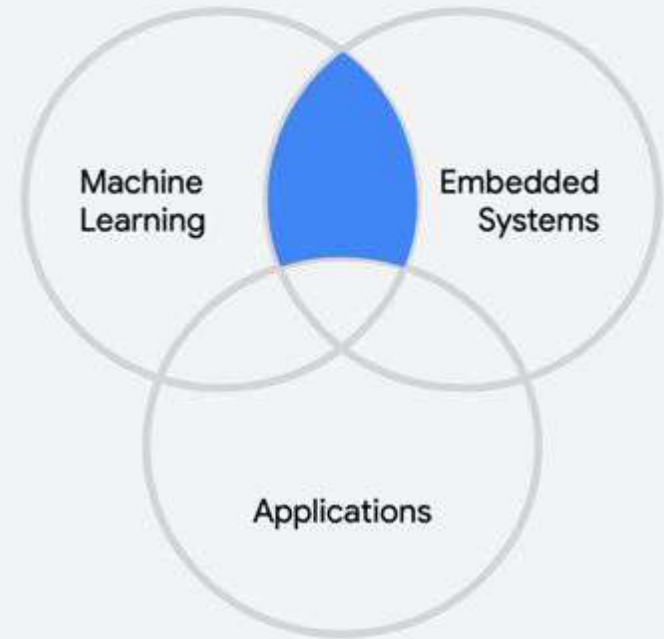
Interactions

How **machine learning** can
enable new and interesting
TinyML applications?



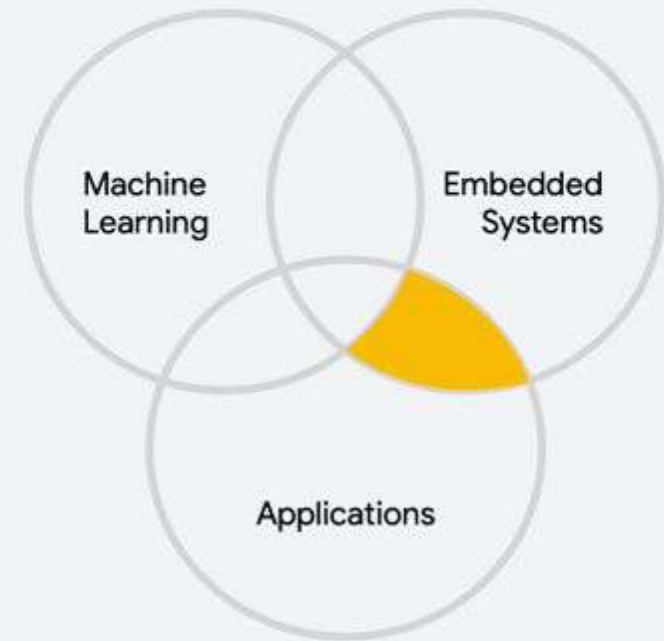
Interactions

What are the **challenges** with enabling **machine learning** on **tiny**, resource-constrained **embedded devices**?



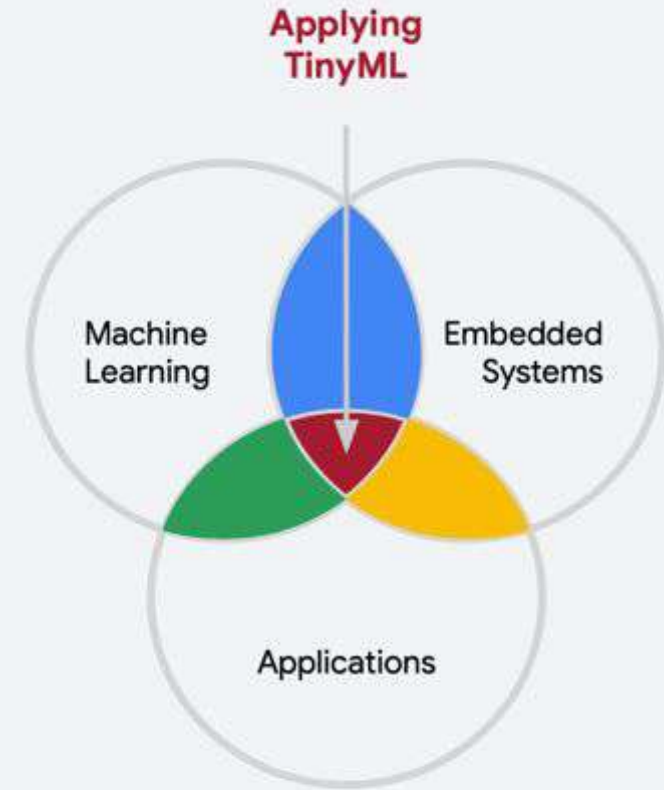
Interactions

What type of new **use cases** can we possibly enable on **embedded systems** that we could not otherwise do before?



At the End of the Day

Given your understanding of things at these various intersections, you will have a deep understanding for **how to apply TinyML**



How are we going to get there?

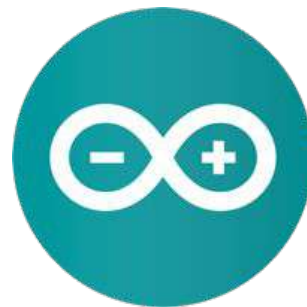
Hands-on Learning

- Software

- Machine Learning
(TensorFlow / LiteRT)
- Programming environments
(Google Colab or Jupyter)
- Train & Deploy Platforms
(Edge Impulse Studio and
SenseCraft AI)

- Hardware

- Arduino Nano 33 BLE Sense
- ESP32S3
- Sensors



TinyML Arduino Kit



XIAOML Kit



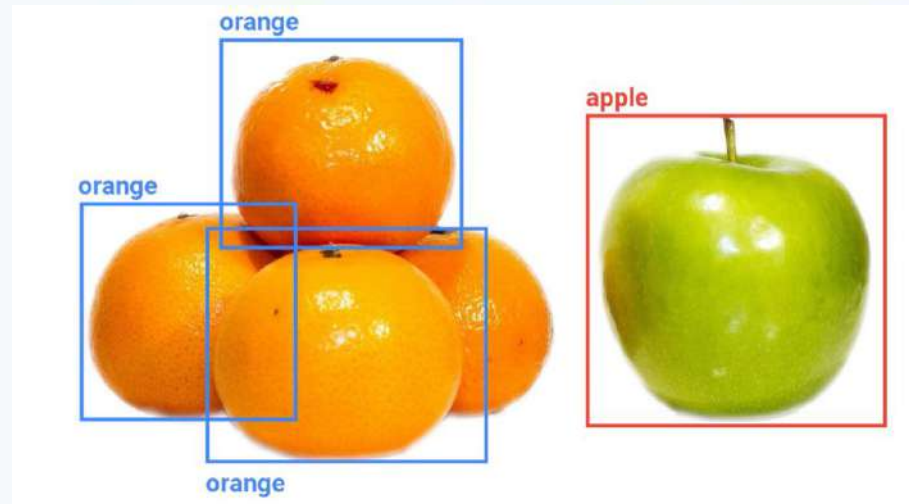
Hands-on Activities

Speech

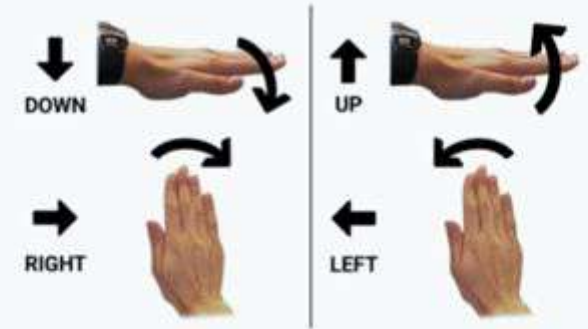


Okay, Google.

Vision



IMU



+



+



+



How is the course structured?

Course Structure

- Weekly video-recorded lectures
 - Hands-on coding / Labs
- Weekly Additional Readings
- Assignments
 - Quizzes (Weekly)
 - Notebooks with codes (4)
 - Hands-on lab reports (4)
- Final Project (Groups of 3 or 4 students)
 - Report
 - Presentation

The screenshot shows a Moodle course page for 'moodle.unifei.edu.br'. The course title is 'iesti01_2026_s1_t01 - TINYML - APRENDIZADO DE MÁQUINA APLICADO PARA DISPOSITIVOS IOT EMBARCADOS'. The page layout includes a left sidebar with a course menu, a main content area with a course introduction and a list of resources, and a right sidebar with a calendar and upcoming events.

Course Menu (Left Sidebar):

- ▼ IESTI01 - TinyML
 - Avisos
 - ▼ Pre-Course Review:
 - Installing Jupyter Notebook & u...
 - Python4DS (Python for Data S...
 - A Whirlwind Tour of Python by ...
 - Main Python Libraries cheatshe...
 - List 1 - Python4DS
 - Forum - Pre-Course
 - ▼ Part 1: Fundamentals
 - ▼ Class 1
 - Slides
 - Video
 - Course Introduction - Slides (A...
 - Pre course survey
 - Forum - Class 1
 - Class 2

Main Content Area:

iesti01_2026_s1_t01 - TINYML - APRENDIZADO DE MÁQUINA APLICADO PARA DISPOSITIVOS IOT EMBARCADOS

Curso | Configurações | Participantes | Notas | Relatórios | Mais ▼

▼ IESTI01 - TinyML [Contrair tudo](#)

The main content area features a large illustration of a bearded man with a red beard and glasses, wearing a red shirt, sitting at a desk with a laptop. He is surrounded by four students (three girls and one boy) who are also working on laptops. The desk is cluttered with various electronic components, including a drone, a robot car, and a small robot. The background shows a workshop or lab setting with shelves filled with electronic equipment and a window with a plant.

Right Sidebar:

Calendário

março 2026

Dom	Seg	Ter	Qua	Qui	Sex	Sáb
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

[Calendário do Curso](#)
[Gerenciar assinaturas](#)

Próximos eventos

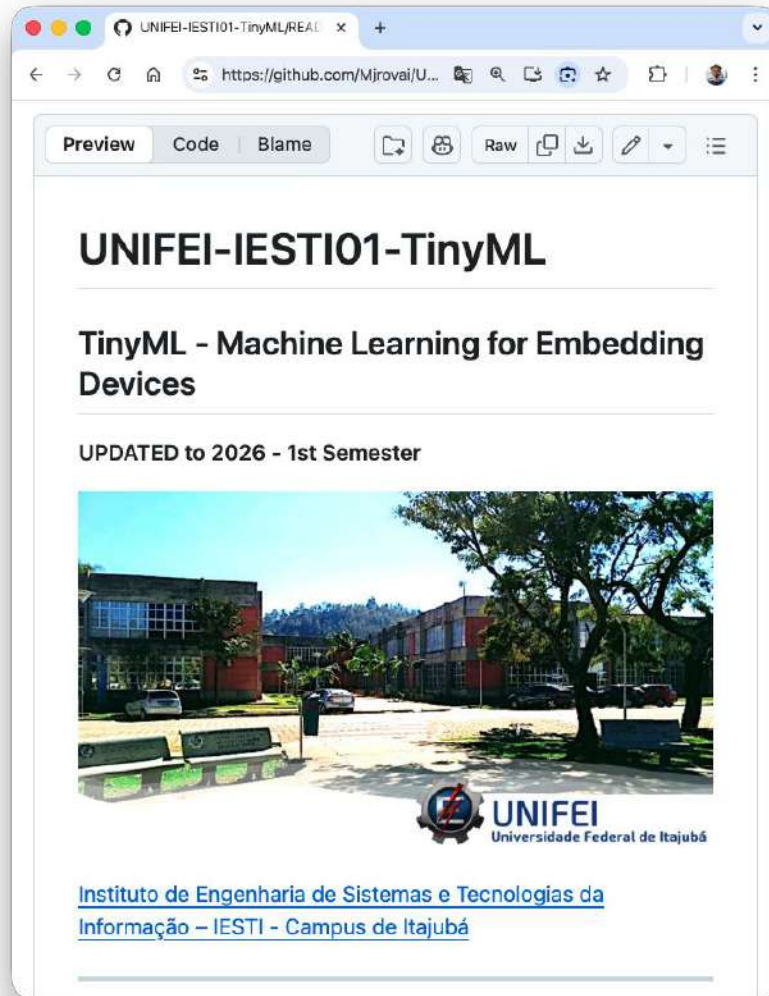
- [Pre course survey está ma...](#)
terça-feira, 17 março, 23:59 PM
- [Quiz 1 está marcado\(a\) pa...](#)
terça-feira, 17 março, 23:59 PM
- [Quiz 2 está marcado\(a\) pa...](#)

Class planning and approval process

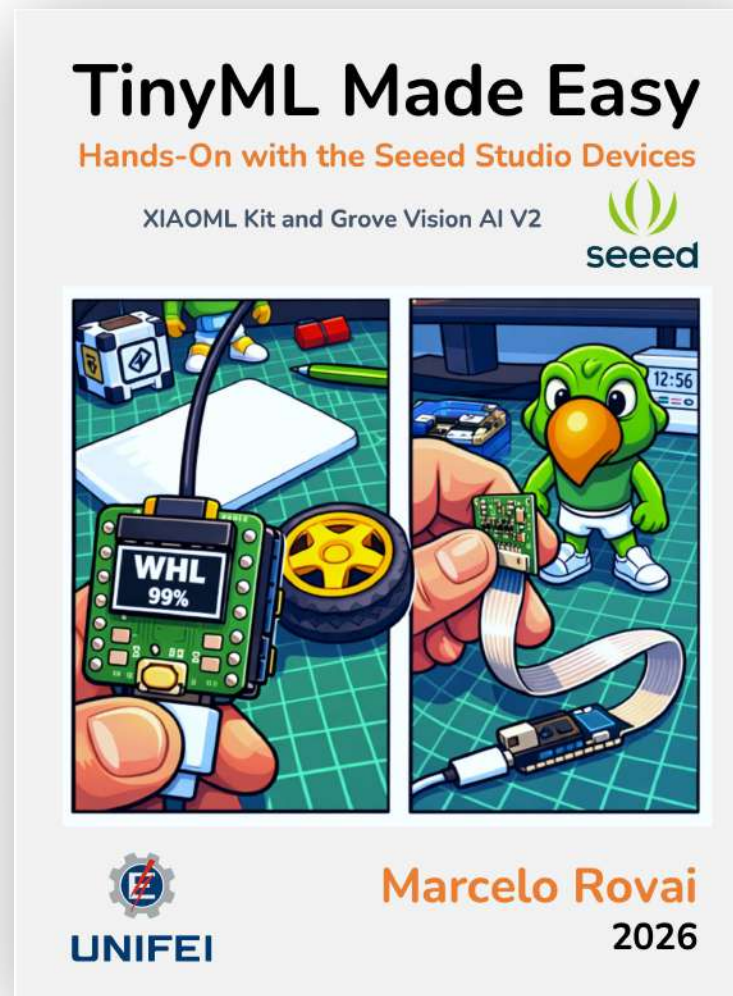
- Minimal suggested Workload (4 hours per week):
 - 30 hours (Weekly recorded classes of about 2h, for 15 weeks)
 - 15 hours of assignments/coding/labs
 - 15 hours in research, individual studies, and final project (in a group)
- Approval process:
 - 1st Evaluation:
 - Individual **Quizzes**: 10%
 - Individual **Exercise Lists** (Notebooks): 25%
 - Group **Project Proposal**: 15%
 - 2nd Evaluation
 - Individual **Quizzes**: 10%
 - Individual **Practical Projects** (Lab reports): 25%
 - Group **Project Presentation (*)** and Final Report: 15%

(*) Examples: IESTI01 – TinyML - Final Projects

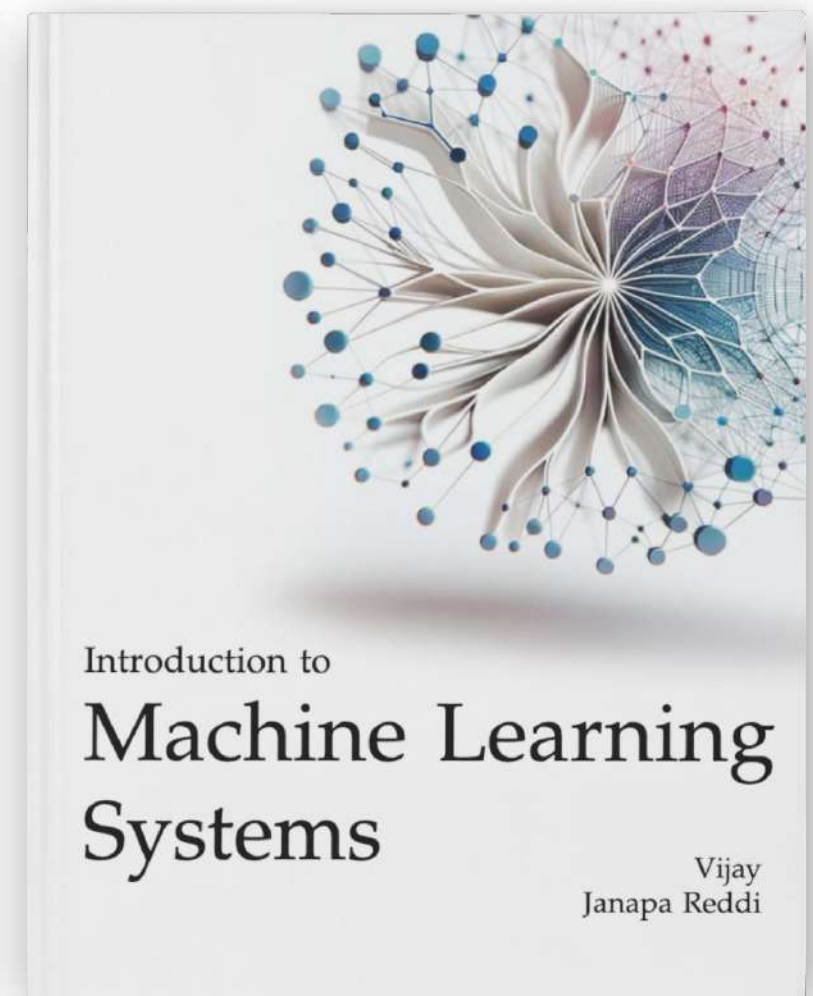
Additional Reading Support



[IESTI01 - Repo](#)



[XIAOML Kit Labs](#)



[MLSys Book](#)

To learn more ...

Online Courses

[Harvard School of Engineering and Applied Sciences - CS249r: Tiny Machine Learning](#)
[Professional Certificate in Tiny Machine Learning \(TinyML\) – edX/Harvard](#)
[Introduction to Embedded Machine Learning - Coursera/Edge Impulse](#)
[Computer Vision with Embedded Machine Learning - Coursera/Edge Impulse](#)
[UNIFEI-IESTIO1 TinyML: “Machine Learning for Embedding Devices”](#)

Books

[“Python for Data Analysis” by Wes McKinney](#)
[“Deep Learning with Python” by François Chollet - GitHub Notebooks](#)
[“TinyML” by Pete Warden and Daniel Situnayake](#)
[“TinyML Cookbook 2nd Edition” by Gian Marco Iodice](#)
[“Technical Strategy for AI Engineers, In the Era of Deep Learning” by Andrew Ng](#)
[“AI at the Edge” book by Daniel Situnayake and Jenny Plunkett](#)
[“XIAO: Big Power, Small Board” by Lei Feng and Marcelo Rovai](#)
[“TinyML Made Easy” by Marcelo Rovai](#)
[“Machine Learning Systems” by Vijay Janapa Reddi](#)
["Tiny Machine Learning Quickstart" by Simone Salerno](#)
[Projects Repository Edge Impulse Expert Network](#)

On the **TinyML4D website**, you can find lots of educational materials on TinyML. They are all free and open-source for educational uses – we ask that if you use the material, please cite them! TinyML4D is an initiative to make TinyML education available to everyone globally.

Thanks



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